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BALNet: Balanced Attack learning Network for Web Attack Detection via Adaptive Sampling, LDA, and CNNs

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Abstract

The swift expansion of interconnected infrastructures including Internet of Things (IoT) ecosystems, cloud computing platforms, and cyber-physical systems has markedly heightened susceptibility to advanced web threats such as SQL injection, cross-site scripting (XSS), and brute-force assaults, which persist in dominating the OWASP 2025 risk framework. Despite the fact that current intrusion detection systems (IDS), incorporating deep learning (DL)-based methodologies, report accuracies surpassing 99% on established benchmark datasets like CIC-IDS2017, their efficacy is frequently undermined by pronounced class imbalance and elevated feature dimensionality. Specifically, minority attack classes, which may represent less than 5% of the dataset, experience inadequate recall, thereby obscuring vulnerabilities linked to high-impact yet infrequent threats. In order to mitigate these constraints, this study presents BALanced Attack Learning NETwork (BALNet), a robust intrusion detection framework that integrates Adaptive Synthetic Sampling (ADASYN) for the precise balancing of sparse minority regions, Linear Discriminant Analysis (LDA) for effective dimensionality reduction, and Convolutional Neural Network (CNN) for the classification of multi-class attacks. Thorough ablation studies conducted on the CIC-IDS2017 dataset evaluate ADASYN against alternative resampling methodologies (such as Synthetic Minority Optimization Technique (SMOTE), Random Over-Sampling (ROS), and Random Under-Sampling (RUS), LDA in comparison to other feature extraction methods (including Principal Component Analysis (PCA), and the evaluation of classification models CNN versus Deep Neural Networks (DNN) and Long Short Term Memory (LSTM) classifiers. The experimental findings reveal that BALNet attains a training accuracy of 99.87%, a test accuracy of 99.79%, and a validation accuracy of 97.31%, with negligible overfitting (training and validation losses recorded at 0.0175 and 0.118, respectively), while significantly enhancing generalization particularly for severely underrepresented attack types such as Web and SQL injection. BALNet establishes a reproducible and scalable IDS benchmark that comprehensively addresses issues of class imbalance and feature redundancy, thus facilitating robust and real-time detection of intricate web attacks within contemporary network environments.

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Keywords Anomaly detection, Web-based attacks, Deep learning, Network security, Intrusion detection system

1 Introduction

The widespread adoption of smartphones, Internet of Things (IoT) devices, and cloud-based platforms has significantly increased the interconnected technologies including the scale, complexity, and vulnerability of modern networks. While ubiquitous connectivity enables seamless data exchange, it exposes network infrastructures and web applications to malicious attacks. Intrusion Detection Systems (IDSs) play a vital role in monitoring network traffic, identifying malicious behavior, and generating alerts for potential cyberattacks.

Early IDS, originating from rule-based and signature-matching methodologies, face challenges against advanced zero-day threats due to their inability to adapt to evolving attack patterns and high-dimensional traffic data traffic data [6]. Moving on, IDSs evolved into diverse frameworks that can be categorized based on deployment (host-based, network-based, node-based, hybrid) and detection strategy. Numerous algorithms based on distance, density [7], kernel, and clustering methods [10] have been proposed within the field of anomaly detection. Rule-based [24] attack prevention and detection systems struggle to effectively counter unknown attacks and are straightforward to detect using predefined rules. These approaches are often limited in their ability to detect novel and evolving attacks.

Over the years, diverse methodologies based on event processing, data mining, statistical and payload modeling, rule-based systems, machine learning, and deep learning have been employed. [3, 12, 22, 27]. Deep models offer the ability to automatically learn hierarchical and complex feature representations from raw network traffic, enabling improved detection of both known and unseen attacks. Frameworks such as Convolutional Neural Networks (CNNs), Long Short-Term Memory (LSTM) networks, and Deep Neural Networks (DNNs) have proven promising results across detection benchmarks. Nevertheless, despite these advances, several critical challenges remain unresolved.

1.1 Research gap and motivation

Although deep learning based intrusion detection systems (IDS) report very high overall accuracies (>99%) on benchmark datasets such as CIC-IDS 2017 and 2018, these metrics can be misleading. Robust performance on majority classes often masks severe detection failures for minority yet high-impact web attacks, raising concerns about the practical reliability of such systems in real-world deployment scenarios. Critical web attack categories exhibit extremely low recall despite contributing minimally to aggregate accuracy figures.

For instance, CIC-IDS 2017 [9] dataset has certain attack categories including Cross-Site Scripting (XSS) and SQL Injection which possess a limited number of samples, thereby leading to highly skewed class distributions. Conventional oversampling techniques typically replicate minority samples or generate synthetic data uniformly, which does not adequately enrich sparse and complex decision regions. Although ADASYN has been proposed to address this issue by selectively generating samples in regions of

high classification difficulty, its application has largely been examined in isolation, without integration into a broader feature-learning and classification framework.

Moreover, modern intrusion detection datasets are characterized not only by class imbalance but also by high feature redundancy and evolving multi-class attack behaviors. Many existing IDS frameworks implicitly refine majority-class discrimination, leading to inflated accuracy of metrics while underperforming rare yet security-critical attack classes. This underscores the need for a unified and systematic detection framework that addresses data imbalance, feature separability, and classification robustness in a holistic manner.

1.1.1 Problem statement

Contemporary DL–IDS, while adept at hierarchical feature learning from raw traffic [27], face two intertwined challenges: (1) severe class imbalance, where rare web attacks constitute <5% of traffic in benchmarks like CIC-IDS 2017 [9], biasing models toward majority (benign) classes and yielding poor minority recall,and (2) high-dimensionality of datasets hindering discriminative pattern extraction without manual intervention [5]. Existing solutions apply balancing (Random Sampling/SMOTE/ADASYN) and extraction techniques like (PCA/LDA) in isolation, lacking systematic pipelines to optimize interplay.

1.2 Contribution of the paper

This study bridges key gaps by proposing these contributions.

- *Proposing BALNet*: A BALanced Attack Learning NETwork, an integrated pipeline optimized for severe imbalances in CIC-IDS 2017 [9] web attacks.
- *Conducting comprehensive ablations*: Systematically assessing balancing techniques (ADASYN, SMOTE, random oversampling, random undersampling) and feature extraction methods (LDA, PCA), alongside classification algorithms, thereby quantifying the trade-offs associated with data integrity, computational efficiency, and minority-class representation.
- *Designing an end-to-end preprocessing pipeline*: Seamless integration of data cleaning techniques, along with balanced and engineered data utilizing hyperparameter-optimized deep learning architectures, wherein the CNN exhibits superior performance metrics, indicating 99.79% test accuracy, 97.31% validation accuracy, and a consistent convergence, demonstrating exceptional efficacy on infrequent attack types.

These contributions enable reliable multi-class detection, mitigating majority bias and high dimensionality for practical deployment.

1.3 Organization of the paper

The subsequent segments of this manuscript are structured as follows: Section 2 reviews the related work, discusses existing limitations, followed by the research gaps and motivation for the proposed study. Section 3 describes the proposed methodology, detailing the overall framework and underlying techniques. Section 4 presents the experimentation and evaluation, including ablation studies Sect. 5 presents results, performance

analysis and convergence behaviour. Section 6 concludes the paper and outlines possible directions for future research, followed by the references.

2 Related works

The segment highlights the prior relevant research that has been undertaken in the field of intrusion detection, emphasizing the significance of class imbalance and feature extraction methodologies, and its influence on deep learning based classification models in web attack detection. Various ensembles have been observed in the context, Liu et al. [15] introduces a network intrusion detection system that addresses the challenges of class imbalance in intrusion data and high computational cost associated with deep learning based detectors. In order to mitigate the challenge of class imbalance, the study employs Adaptive Synthetic Sampling (ADASYN), which dynamically generates synthetic samples of minority classes. For the classification task, the researchers utilize LightGBM, an ensemble model based on gradient-boosted decision trees. The findings indicate that the integration of ADASYN and LightGBM significantly enhances the detection of minority attacks while achieving an overall accuracy rate of up to 99.91% on CICIDS2017, 92.57% on NSL-KDD, and 89.56% on UNSW-NB15 datasets.

Yasotha et al. [25] focus on enhancing intrusion detection within Industrial Internet of Things (IIoT) networks, wherein extensive connectivity and high dimensional data generation heighten susceptibility to cyberattacks. The authors propose a wrapper-based feature selection methodology integrated with Direct Linear Discriminant Analysis (WDLDA), in conjunction with a deep learning framework, to diminish redundant features and optimize detection efficiency. When evaluated on the NSL-KDD dataset, the proposed methodology achieves a detection accuracy of up to 97.71%, exhibits a low false positive rate, and entails reduced computational and training time relative to existing approaches. The findings illustrate that the amalgamation of feature selection with LDA-based classification and optimization techniques substantially enhances IDS performance in IIoT contexts.

Balla et al. [4] investigates the effect of class imbalance on deep learning–driven intrusion detection systems (IDSs) within IoT-enabled SCADA environments by using two imbalanced datasets, the Morris power dataset and CICIDS2017. A CNN - LSTM binary classification model is assessed in combination with data balancing techniques, including random sampling, one-sided selection, near-miss, SMOTE, and ADASYN. The experimental findings reveal that addressing class imbalance improves IDS performance, both undersampling and oversampling contribute positively to detection accuracy and F1-score, while hybrid sampling leads to performance degradation attributed to data distortion. Oversampling achieves the highest accuracy of 99.47%, recall of 99.49%, and F1-score of 99.46%, with undersampling closely following. In conclusion, the results underscore the pivotal importance of judiciously selecting data balancing strategies, particularly oversampling, in the development of robust IDSs for imbalanced intrusion datasets.

Zakariah et al. [26] proposes a framework for intrusion detection in IOT networks, with machine learning, addressing two significant challenges: class imbalance and suboptimal detection rates for minority attack vectors. The authors integrate Adaptive Synthetic Sampling (ADASYN) with deep learning models, including CNN–LSTM with an attention mechanism and Multi-Layer Perceptron (MLP) classifiers, to enhance rare

attack detection in the NSL-KDD dataset. ADASYN enhances the detection capabilities for challenging attack categories, namely Remote-to-Local (R2L) and User-to-Root (U2R). Empirical findings substantiate that the proposed hybrid models attain superior accuracy, F1-score, and Area Under the Curve (AUC) metrics than traditional ML and comparable deep learning approaches, with the most effective models achieving accuracy levels exceeding 89% accuracy and F1-score and AUC values as high as 0.94. This research validates the efficacy of integrating ADASYN with deep learning frameworks while highlighting prospective directions including enhanced detection of minority attacks, multi-class classification challenges, and advancements in model interpretability.

Liu and Guo [16] investigate Weighted Linear Discriminant Analysis (WLDA), which serves as a cost-sensitive augmentation of conventional Linear Discriminant Analysis (LDA) aimed at mitigating performance issues degradation from class imbalance in high-dimensional supervised feature extraction tasks. In this study, weights that are inversely proportional to class sample sizes are allocated to ensure enhanced class separability, allowing minority classes to contribute more significantly to the learned discriminant space. The authors propose a solution algorithm for WLDA, scrutinize its computational complexity, and empirically demonstrate that WLDA consistently surpasses standard LDA as well as other comparative methodologies when applied to imbalanced datasets.

Sarhan et al. [20] examine the generalizability of feature extraction (FE) and machine learning methodologies for Network Intrusion Detection Systems (NIDSs) within IoT environments. The research assesses three FE techniques including PCA, Autoencoders, and LDA, in conjunction with six ML models including Deep Feedforward (DFF), CNN, Recurrent Neural Networks (RNN), Decision Trees (DT), Logistic Regression (LR), and Naïve Bayes (NB) across three established benchmark datasets: UNSW-NB15, ToN-IoT, and CSE-CIC-IDS2018. The results indicate that no singular FE-ML combination consistently outperforms others across all datasets, revealing that model efficacy is highly contingent upon the dataset utilized. The paper concludes that enhancing IDS performance by fine-tuning models for specific datasets possesses limited practical value, advocating instead for the development of generic benchmark feature sets, improved generalizability, and increased incorporation of explainable artificial intelligence (XAI).

Abdullah et al. [1] addresses key challenges in Software Defect Prediction (SDP) and intrusion detection by emphasizing the criticality of effective feature selection and extraction within high-dimensional, intricate datasets. The authors propose Depth Adaptive Sine Cosine Feature Selection (DASC-FS), which integrates an enhanced metaheuristic search algorithm with a robust discriminant method, Depth Linear Discriminant Analysis (D-LDA), by incorporating matrix depth concepts to handle outliers and overlapping classes robustly. In summary, the study emphasizes the significance of sophisticated, depth-aware discriminant analysis and adaptive search strategies in augmenting prediction accuracy, minimizing dimensionality, and improving interpretability.

Alharbi [2] investigates the effectiveness of resampling techniques in addressing class imbalance within IoT intrusion detection, utilizing the IoT_Modbus multiclass dataset as a benchmark. The authors evaluate ensemble classifiers Random Forest (RF), Gradient Boosting, XGBoost, LightGBM, and CatBoost in combination with SMOTE and ADASYN to improve the detection of minority cyber threats. Experimental outcomes indicate that Random Forest achieves superior overall performance, reaching 99.06% accuracy with SMOTE and 99.00% with ADASYN, thereby demonstrating strong

Table 1 Overview of related deep learning-based IDS studies

S. no.	Focus area	Architecture	Dataset	Results	Shortcomings	Comparison with BALNET (proposed)
1. [2]	Class imbalance handling in IoT intrusion detection	SMOTE/ ADASYN + Ensemble classifiers (RF, XGBoost, LightGBM, CatBoost)	IoT_Modbus	RF achieved 99.06% (SMOTE) and 99.00% (ADASYN) accuracy	Limited to ML ensembles; no deep learning or feature learning considered	BALNet offers a deep learning framework for multi-class web attack detection, facilitating enhanced feature learning and minority-class detection with recall range up to 0.9
2. [16]	Cost-sensitive feature extraction for imbalanced data	Weighted Linear Discriminant Analysis (WLDA)	NSL-KDD, UNSW-NB15, CICIDS2017	WLDA outperform standard LDA in class separability	Manual Feature Engineering	The existing study focuses on improving feature extraction via WLDA for imbalanced classification, whereas proposed work integrates aims to achieve end-to-end IDS performance
3. [1]	Feature selection and extraction	DASC-FS with Depth LDA (D-LDA), PCA, AE	SDP datasets -JM1, KC1, and PC1	Improved robustness, interpretability, and prediction accuracy	Though it present strong findings but study focuses on software defect prediction (SDP) rather than network intrusion	Feature selection in software defect prediction is overridden by focusing on combining imbalance handling and dimensional reduction for improved minority-class detection
4. [20]	Generalisability of FE–ML combinations for NIDS	CA, AE, LDA with CNN, RNN, DT, LR, NB	UNSW-NB15, ToN-IoT, and CSE-CIC-IDS2018	No single FE-ML combination performs well across datasets	Generalization across datasets remains limited	While single FE-ML combination remains unrecognized in the former study, an integrated pipeline is a proposed for CIC IDS [9] dataset, which necessitates validation across multiple datasets
5. [26]	Minority attack detection in IoT IDS	ADASYN + CNN–LSTM with attention, MLP	UNSW-NB15, ToN-IoT, CSE-CIC-IDS2018	No single FE–ML pair consistently superior; dataset dependency observed	Lack of a generic feature representation; limited real-world deployment insight	The referenced study indicates accuracy under 90%. In contrast, BALNet exhibits mean accuracy exceeding 99%, reduced variance across folds
6. [18]	Multi-class Network Intrusion Detection with Class Imbalance via LSTM and SMOTE	LSTM with SMOTE oversampling and categorical focal loss for class imbalance	KDD 99 and CIC [9]	Achieved 99.33% overall accuracy and 91.95% recall, with Web Attack detection reaching 93% F1-score (grouped XSS/SQL), PPV of 92%, though precision on minority classes remained comparatively lower	Limited handling of extreme minority classes and potential overfitting due to sequential model assumptions	Although high accuracy was attained, the model's use of SMOTE and grouped classes diminishes precision for minority attacks. In contrast, BALNet's ADASYN-LDA pipeline enhances class separability and bolsters generalization for minority classes

Table 1 (continued)

S. no.	Focus area	Architecture	Dataset	Results	Shortcomings	Comparison with BALNET (proposed)
7. [4]	Impact of imbalance on DL-based SCADA IDS	CNN–LSTM with sampling techniques	NSL-KDD	NSL-KDD, UNSW-NB15, CICIDS2017	Hybrid sampling caused data distortion; limited model diversity	The study assesses class imbalance in SCADA IDS using CNN–LSTM, noting data distortion from hybrid sampling, while BALNet utilizes ADASYN with LDA and CNN, enhancing minority class detection
8. [25]	Feature reduction for IIoT intrusion detection	Wrapper-based FS + WDLDA with DL	Morris Power, CICIDS2017	Modified MNIST, Butterfly images	Evaluated on a single dataset; limited scalability analysis	The study applies wrapper-based feature selection with WDLDA for IIoT IDSt, whereas BALNet enables scalable, multi-class intrusion detection through extensive studies
9. [15]	Class imbalance and efficiency in NIDS	ADASYN + LightGBM	NSL-KDD	Accuracy up to 99.91% on CICIDS2017	Relies on ML models; lacks deep feature learning	The study addresses class imbalance via ADASYN and LightGBM, achieving notable accuracy, but BALNet works with promising DL tools, along with feature extraction for multi-class attack detection
10. [8]	Supervised dimensionality reduction using DL features	CNN feature extraction + LDA	Modified MNIST, Butterfly images	Comparable to supervised UMAP and ButterflyNet	Not designed for anomaly or intrusion detection	Reference showcases supervised dimensionality reduction through LDA on CNN features, while BALNet incorporates LDA in a comprehensive IDS framework (ADASYN + LDA + CNN) for mitigating class imbalance and high-dimensional data challenges

robustness to imbalanced data. The findings suggest that SMOTE and ADASYN yield comparable enhancements, and that ensemble methodologies exhibit greater resilience to data imbalance compared to single classifiers.

In our preceding investigation [13], we examined the role of deep learning methodologies for network anomaly detection, driven by the rapid growth of connected devices that generate substantial data requiring efficient computation and transmission. In contrast to conventional machine learning paradigms, deep learning models exhibit a markedly enhanced capacity for automatic feature extraction, elevated detection precision, and the capability to apprehend intricate non-linear patterns that bear resemblance to cognitive learning mechanisms. The review highlights persistent challenges such as feature selection, zero day threat detection, model interpretability, and high computational overhead. Overall, this prior work established deep learning as a promising yet evolving

paradigm and laid the foundation for subsequent research aimed at development of more accurate, resilient, and interpretable cybersecurity solutions.

Several studies in literature have explored the integration of statistical techniques with deep feature representations, extending beyond the direct purview of anomaly detection. Notably, prior work [8] investigated the application of Linear Discriminant Analysis (LDA) on features extracted from various layers of pretrained convolutional neural networks (CNNs) for supervised dimensionality reduction. In this approach, LDA is employed to project high-dimensional CNN features into a lower-dimensional space while preserving global class structure, resulting in embeddings where semantically similar classes are positioned closer together. Experimental validation on a modified MNIST dataset and a biological image dataset of butterfly subspecies demonstrated performance comparable to state-of-the-art supervised embedding methods. The approach harnesses explicit class-based constraints via LDA, providing enhanced interpretability and greater control over the embedding landscape. This research highlights a significant trend in deep learning towards hybrid frameworks that combine unsupervised feature extraction with supervised learning. Such integrations are particularly relevant for anomaly detection, emphasizing the importance of low-dimensional representations and class separability. The evaluation of related works is summarized in Table 1.

3 Proposed methodology

The methodological framework in this study is structured, multi-phase, aimed to effectively detect web-based network outliers using deep learning techniques. CIC-IDS 2017 [9] is a collection of high-quality, labeled, repository of standardized network traffic data, renowned for its comprehensive representation of realistic traffic and attack. To ensure consistency and relevance, the raw data undergoes a thorough preprocessing phase,missing values are addressed, categorical variables are encoded, and feature scaling is implemented to ensure homogeneity. An analysis of class imbalance within the dataset is conducted, wherein various balancing methodologies such as Random Oversampling, Random Undersampling, SMOTE, SMOTEEN, and ADASYN are examined to accommodate instances of the minority class and to augment detection sensitivity. This strategic approach enhances the generalization capabilities of the model and increases its sensitivity to rare yet critical attack patterns within cybersecurity datasets [14].

The processed and balanced dataset is subsequently assessed for feature engineering using PCA, and LDA [11, 19] and analyzed through deep learning architectures, CNN, DNN, and LSTM, each trained and refined to capture diverse feature patterns and temporal dynamics inherent to the traffic data. The efficacy of the models is assessed through multiple metrics including accuracy, precision, recall and F1-score complemented by cross-validation techniques to guarantee robustness and generalizability. It is imperative to note that a high aggregate accuracy metric does not adequately reflect the operational effectiveness of an IDS particularly in the presence of severely imbalanced datasets and critical web attack classes. This end-to-end methodology provides a comprehensive framework for anomaly detection in web traffic, balancing performance, efficiency, and real-world applicability.

Figure 1 illustrates the proposed algorithm from raw data ingestion to final classification.

```
# Phase 1–2: Data Collection & Preprocessing
D ← load_dataset(path, encoding = cp1252)
D ← normalize_feature_names(D)
D ← replace_nan_inf(D)
y ← label_encode(D.labels)
X_train, X_test, y_train, y_test ← stratified_split(D.features, y)
X_train ← standard_scale(X_train)
X_test ← apply_scaler(X_test)

# Phase 3: Data Balancing (Training Set Only)
for sampler ∈ {ROS, RUS, SMOTE, SMOTEENN, ADASYN} do
    X_s, y_s ← sampler.fit_resample(X_train, y_train)
    record(sampler, runtime, class_distribution(X_s))
end for
X_bal, y_bal ← select_sampler(ADASYN)

# Phase 4: Feature Engineering & Extraction
R ← compute_pearson_correlation(X_bal)
X_bal, X_test ← remove_correlated_features(X_bal, X_test, R)

for extractor ∈ {PCA, LDA} do
    if extractor == PCA then
        Z_train, Z_test ← PCA.fit_transform(X_bal, X_test)
    else if extractor == LDA then
        Z_train, Z_test ← LDA.fit_transform(X_bal, y_bal, X_test)
    end if
    score(extractor) ← evaluate(Z_train, y_bal,
                                metrics = {Accuracy, Precision, Recall, F1})
end for
Z_train, Z_test ← select_extractor(LDA)

# Phase 5: Deep Learning Classification
for model ∈ {CNN, DNN, LSTM} do
    M ← initialize(model)
    M.train(Z_train, y_bal)
    performance(model) ← cross_validate(M, Z_train, y_bal,
                                        metrics = {Accuracy, Precision, Recall, F1})
end for
M* ← select_model(performance, criterion = F1)

# Phase 6: Final Evaluation
results ← evaluate(M*, Z_test, y_test,
                   metrics = {Accuracy, Precision, Recall, F1})

return M*, results
```

Fig. 1 Algorithm for proposed methodology

3.1 Phase 1: data collection

The CICIDS2017 dataset, developed by the Canadian Institute for Cybersecurity, provides a robust benchmark for research in intrusion detection by emulating realistic network traffic patterns and contemporary cyber threats. It encompasses both benign and malicious behaviors presented in PCAP and CSV files, along with labeled flow data generated using CICFlowMeter. The dataset is constructed utilizing the interactions of 25 synthetic users engaging over prevalent protocols such as HTTP, HTTPS, FTP, SSH, and email, within a meticulously defined network topology. This environment comprises 12 victim machines and a distinct attack network, thereby facilitating the simulation of authentic attack scenarios in conjunction with typical user activity. With an extensive array of over 80 extracted flow features and comprehensive labeling, the dataset is curated and collected for profiling real behavior, catering to diverse protocols, metadata, and attack classes with an aim to satisfy a comprehensive set of criteria necessary for building robust intrusion detection models and includes a complete network configuration with components like modems, firewalls, memory dumps, and system calls, while maintaining high data fidelity and a rich feature set for analysis [21]. The traffic has been

recorded over a span of 5 days, distributed across 5 files and is categorized into 15 distinct classes and encompasses 83 features.

Table 2 illustrates CSV files along with attack profiling for the CIC IDS 2017 [9] dataset.

Given the OWASP 2025 threat landscape, the list enumerates injection attacks and authentication failures, accordingly we target web attacks, including SQL injection, brute force and cross-site scripting, as they are pervasive and consequential with prevailing industry standards. We address these attacks to ensure alignment with fundamental security concerns, relevance, and necessity of detection methods within real-world web protection efforts.

In the context of SQL Injection, attackers inject malicious SQL code into input fields such as login forms and search bars, which subsequently concatenates into database queries undergoing appropriate sanitization, thereby facilitating unauthorized access, modification, or deletion of data. This remains a high-impact vulnerability due to direct database compromise.

XSS is malicious JavaScript that is injected via user inputs and executed when users' encounter compromised content, enabling session hijacking, cookie theft, and phishing. The attack is prioritized as injection-like due to payload execution in a trusted context.

In Brute-Force attackers methodically try credential combinations against login endpoints until success, frequently employing automation via wordlists or dictionary attacks to breach weak passwords, thereby facilitating account takeovers.

3.2 Phase 2: data cleaning and preprocessing

Data cleaning and preprocessing is an essential process prior to integration and analysis. Web Attacks CSV file with cp1252 encoding is done, followed by data cleansing and preprocessing procedure. Column name standardization is performed by converting column names to lowercase and removing spaces and extraneous parentheses. Infinite and NaN values are uniformly replaced with −1. Comparisons for before and after datacleaning are drawn in Sect. 4.

3.2.1 Encoding categorical variable

A Label encoder instance is initialized to transform categorical class labels into integer encoding, as it prepares the classification labels for numerical analysis [17], suitable for machine and deep learning models. The encoder is fitted on training labels (y_train)

Table 2 Attack profiling

Day	CSV file	Network traffic/attack types
Monday	Monday-WorkingHours.pcap_ISCX.csv	Benign
Tuesday	Tuesday-WorkingHours.pcap_ISCX.csv	Brute Force, SFTP, SSH
Wednesday	Wednesday-Workinghours.pcap_ISCX.csv	DoS, Heartbleed
Thursday	Thursday-WorkingHours-Afternoon-Infilteration.pcap_ISCX.csv, Thursday-WorkingHours-Morning-WebAttacks.pcap_ISCX.csv	Web and infiltration attacks
Friday	Friday-WorkingHours-Afternoon-DDos.pcap_ISCX.csv, Friday-WorkingHours-Afternoon-PortScan.pcap_ISCX.csv, Friday-WorkingHours-Morning.pcap_ISCX.csv	DDoS, Botnet, PortScan

during which a deterministic mapping from each unique class label to an integer index is in range [0, n – 1] is learned, where n is the number of attack classes plus one (Benign). The same encoder is subsequently utilized on the test labels to ensure that the test dataset adheres to an identical label-to-index mapping, ensuring consistency for training and evaluation phases.

3.2.2 Dataset splitting

The dataset is partitioned into training and testing subsets for feature vectors (X) and their associated class labels (y), with a random allocation of 80% designated for training and 20% for testing, respectively. A fixed random seed with value 42 is used to ensure reproducibility of the split across experimental iterations. Prior to the division, the data undergoes shuffling to eliminate any potential biases related to ordering.

Subsequent to the split, data balancing techniques are applied to the training subset (X_train, y_train), while the test set is retained to ensure that model evaluation reflects outcomes on previously unseen data. outcomes on previously unseen data. This partitioning strategy guarantees equitable evaluation, reproducibility, and statistical validity, rendering it particularly suitable for imbalanced, multi-class intrusion detection contexts.

3.3 Phase 3: data balancing

Data balancing is essential in intrusion detection systems as real network traffic datasets are highly skewed toward benign behavior, causing learning algorithms to favor majority classes and overlook rare yet critical attacks such as SQL Injection and XSS in our case. This imbalance leads to misleadingly high accuracy while yielding poor minority-class detection and high false-negative rates, which are unacceptable in security-critical environments. By rebalancing class distributions, minority attack regions are represented adequately, decision boundaries become informative, and classifiers can learn discriminative patterns more effectively.

We have used conventional oversampling, undersampling, and hybrid methods for severe class imbalance using the CIC-IDS 2017 [9] dataset. This includes evaluation of oversampling techniques including Random Oversampling (ROS), Synthetic Minority Overampling TEchnique (SMOTE), ADAptive SYNthetic Sampling (ADASYN), undersampling techniques cover Random UnderSampling (RUS) and hybrid techniques like SMOTE + Edited Nearest Neighbour (SMOTEEN).

Table 3 categorizes data balancing methodologies by mechanism and relevance to attack classification.

3.4 Phase 4: feature engineering

In this stage Pearson correlation matrix is constructed and analyzed to assess relationships between functionality and multicollinearity among numerical attributes. Due to the high feature dimensionality, a single heatmap is cluttered and hard to interpret. To enhance clarity, a segmented heatmap approach is employed by dividing the correlation matrix into smaller feature subgroups where each segment's partial correlation matrix is visualized. Correlations with an absolute value above 0.8 are annotated to enhance interpretability, while weaker correlations are suppressed. These are detailed in the experimentation section.

Table 3 Evaluation of different data balancing techniques

Technique	Category	Core mechanism	Network intrusion detection	Web attack classification	Anomaly detection systems
ROS	Oversampling	Random duplication of minority samples	Improves recall for known attacks	Useful for quick baseline evaluation	Limited anomaly generalization
SMOTE	Oversampling	k-NN-based synthetic interpolation	Balanced multi-class attack detection	Rapid prototyping and baseline modeling	Improves learning of known anomalies
ADASYN	Oversampling	Density-based adaptive synthesis	Handles rare and emerging attack types	Generates diverse attack variants	Strong edge-case and sparse anomaly
RUS	Undersampling	Random removal of majority samples	Reduces dominance of benign traffic	Simplifies web attack datasets	Coarse anomaly separation
SMOTEEN	Hybrid	SMOTE + ENN noise removal	High precision training	Reduces false positives	Refined anomaly boundaries

Further for feature engineering, Linear Discriminant Analysis (LDA) and PCA are analyzed. Principal Component Analysis is an unsupervised technique for dimensionality reduction through the projection of data onto orthogonal components. The approach captures maximum variance through the eigenvalue decomposition of the covariance matrix.

LDA projects data into lower dimensions, retaining the most discriminative information between classes by maximizing separations and locating a linear combination of features that best separates class and ensures informative features are obtained to the model. Further an ablation study on deep learning techniques, with and without LDA is done to assess the impact of model speed and efficiency.

LDA seeks projection directions '*w*', maximizing the Fischer criterion: ratio of between-class S_B to within class scatter S_w.

$$\mathrm{J(w)} = \mathrm{w^T S_B w / w^T S_w w}.$$

Solved via generalized eigenvalue problem: $S_B w = \lambda S_W w$, yield eigenvectors up to K – 1 for K classes.

3.5 Phase 5: classification: deep learning models

Undermentioned deep models have been studied and evaluated based on various hyperparameters.

3.5.1 Convolutional neural network (CNN)

CNNs are supervised deep learning architectures that demonstrate the ability to discern spatial hierarchies within datasets.. The framework comprises convolution, layers, and fully connected (dense) layers, which facilitate hierarchical extraction.

Let the input record for a single feature vector be:

$$\mathrm{X} = [\mathrm{x_1}, \mathrm{x_2}, \mathrm{x_3}, \ldots \mathrm{x_n}] \in R^d$$

where $\mathrm{x_i} \in R^d$ represent the ith feature and d is the total number of features. The class label for the record is cl ∈ {1,2,…,C} where C is the number of classes.

A convolution filter $w \in R^{f*d}$ of size f slides across the input feature vector to produce the feature map. The convolution operation for position i is defined as:

$$h_i = \tan h\,(w \cdot x_{i:i+f-1} + b)\,, \quad i = 1, 2 \ldots,\ n - f + 1,\ \text{where}$$

$x_{i:i+f-1}$ is the subset of the input vector covered by the filter at position i
$b \in R$ is the bias term,
$\tanh\boxed{f_0}(\cdot)$ is the activation function.

The complete feature map for a single filter is then:

$$h = [h_1, h_2, \ldots h_{n-f+1}] \in R^{n-f+1}$$

Multiple filters $\{w^{(1)}, w^{(2)},\ldots,w^{(k)}\}$ generate k feature maps. A max-pooling operation is applied to each feature map to reduce dimensionality while retaining the most significant activations:

$$h_j^{pool} = \max\boxed{f_0}\left(h^{(j)}\right),\ j = 1, 2, \ldots, k$$

Finally, the pooled features from all filters are concatenated to form the input for the fully connected (dense) layers:

$$h^{fc} = [h_1^{pool}, h_2^{pool}, \ldots, h_k^{pool}] \in R^k$$

This ensures consistent construction of the feature map across all input features, preserving spatial hierarchies and enabling hierarchical representation learning for classification.

3.5.2 Deep neural network (DNN)

DNN is a distinct neural network characterized by its adherence to a feed-forward network topology. The architecture comprises several fully connected hidden layers, including the input and output layers. The mathematical formulation of can be expressed as: $d \in N$: represent the input dimension, N is the total number of layers, and σ: $R \to R$, a nonlinear activation function, each layer $\ell = 1,\ldots L$ performs an affine transformation.

$$W^l : \mathbb{R}^{Nl-1} \to \mathbb{R}^{Nl},$$

where $\ell = 1, \ldots, L$ represents affine linear mappings $X \to AX + b$.

Consequently, $\phi : \mathbb{R}^d \to \mathbb{R}^{Nl}$ is defined as

$$\phi(x) = W^L \sigma(W^{L-1}\sigma(\ldots\sigma(W^1(x)))),\ \text{where}\ x \in \mathbb{R}^d$$

3.5.3 Long short-term memory (LSTM)

LSTM is primarily used for time series data that uses feedforward and feedback connections, to hold model information. It uses a series of gates including forget, input, and output gate to control information flow in each cell in the architecture. The output of the forget gate is denoted as Ft and WF, UF, bF are weights and bias parameters of the forget gate. It is the output of the forget gate, and WI, UI, and bI are the input gate weight and bias.

During training, the weight and bias parameters are optimized. Let x^t denote the input vector and h^t, the hidden state vector at time *t*. The forget gate F^t, input gate I^t, and output gate O^t are computed as:

$$F^t = \sigma(WFx^t + UFh^t - 1 + bF)$$
$$I^t = \sigma(WIx^t + UIh^t - 1 + bI)$$
$$\mathrm{O}^t = \sigma(W\mathrm{O}x^t + U\mathrm{O}h^t - 1 + bO).$$

The cell state C_t is updated by combining the previous cell state C_{t-1} and the current candidate memory content:

$$\mathrm{C_t} = \tanh(Wcx^t + U_{\mathrm{Cht}-1} + \mathrm{bc}).$$

$$C_t = F_t \odot C_{t-1} + I_t \odot C_t$$

Finally, the hidden state h_t is computed as the element-wise product of the output gate and the activated cell state: $h_t = \mathrm{O}_t \odot \tan\, h(Ct)$.

The evaluation of three models is detailed in the experimentation section.

3.6 Phase 6: model evaluation

This phase encompasses the model evaluation, training strategies, evaluation metrics, and implementation specifics employed to ensure robust, reliable, and reproducible experimental results.

4 Experimental setup and implementation details

The experimentation is executed utilizing the Python framework, using open-source libraries for machine and deep learning workflows. The deep learning models are designed, trained, and evaluated through the TensorFlow and Keras frameworks owing to their adaptability, robustness, community support, and seamless integration with supplementary Python-based tools such as pandas for data cleansing and manipulation, NumPy for numerical analyses, and Scikit for visual data representations. The graphical outputs are produced using matplotlib and seaborn, which facilitate the creation of comprehensive visualizations including heatmaps, class distributions, and training performance curves.

Hyperparameter optimization encompassing learning rate, dropout rates, neuron counts, and network depth is executed through a synthesis of manual adjustment and grid search, informed by trends in validation accuracy and loss. Various optimizers including Adam, RMSprop, and Stochastic Gradient Descent (SGD) are critically assessed, with the optimal configuration determined based on the stability and convergence efficiency of the model.

4.1 Evaluation on CIC IDS 2017 [9] dataset

The evaluation of CIC IDS 2017 [9] is detailed in the undermentioned phase.

4.1.1 Dataset analysis

The CIC-IDS 2017 [9] 'Thursday Web and Infiltration Attack' dataset used for the web attack detection is presented in CSV format, encompassing a total of 170,366 rows and 80 columns. Fundamental characteristics of the set are analyzed for its structure,

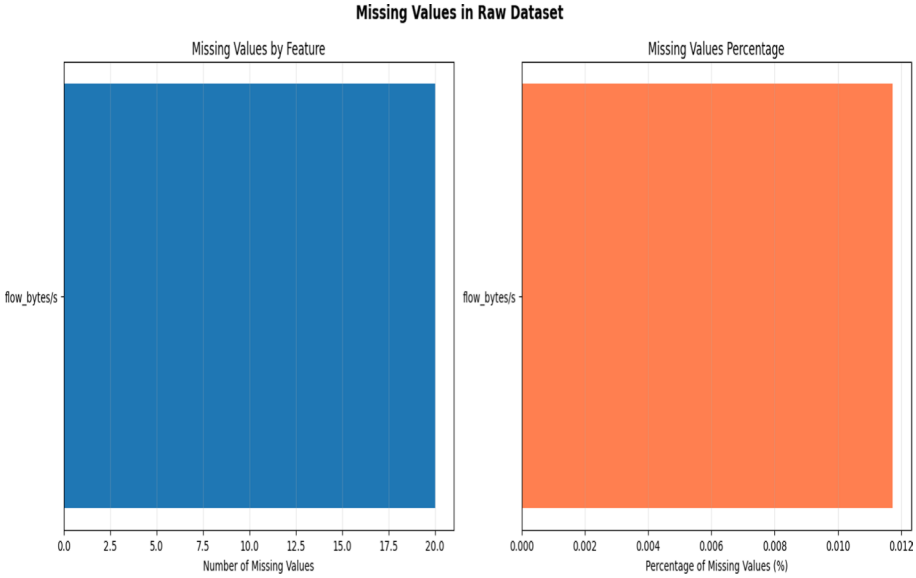


Fig. 2 Missing values in raw dataset

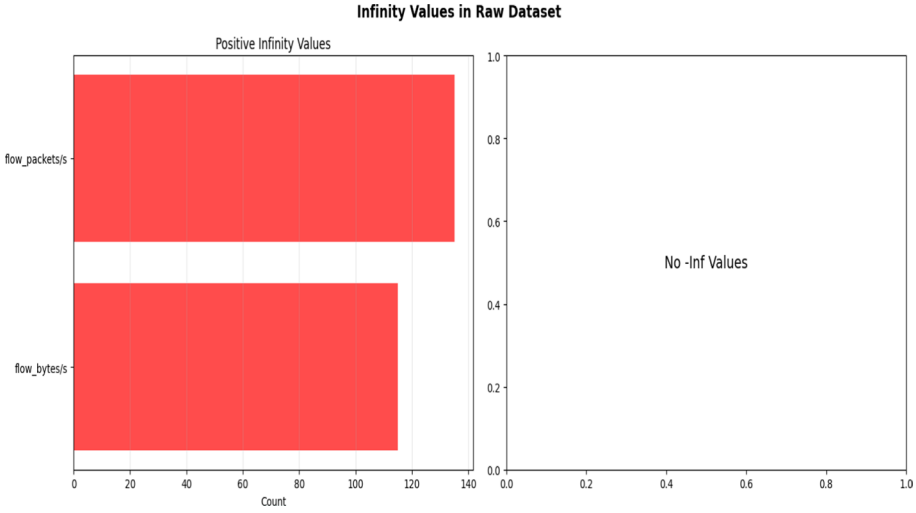


Fig. 3 Infinity values in raw dataset

memory consumption, and data types. Furthermore evaluation of missing infinite values, revealing approximately 20 missing entries and 250 infinite values within the web attacks dataset.

Figure 2 represents an analysis of missing values across the dataset, which frequently originate from issues related to packet capture, incomplete flow records, or error parsing during feature extraction. Untreated, it can distort statistical properties and degrade model performance. Figure 3 depicts infinite values in the repository, which can disrupt normalization procedures and learning algorithms. These are handled through replacement, clipping, and normalization to guarantee numerical stability and the reliability of model training.

Data cleaning procedure of raw network traffic is illustrated in Fig. 4, showcasing the dataset pre and post-preprocessing. The initial analysis highlights the presence of missing, infinite, and inconsistent values and the following graph demonstrates the success



Fig. 4 Before vs. after data cleaning

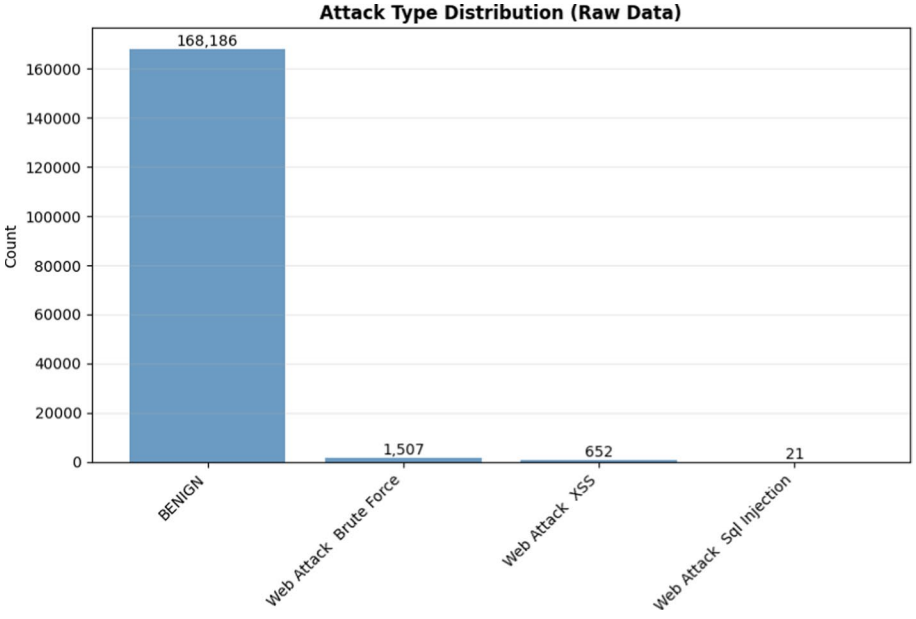


Fig. 5 Attack type distribution

of the data cleaning process, as missing and infinite values are corrected, invalid records are either rectified or removed, and features are standardized for usability.

The dataset contains four distinct labels: Web Attack–Brute Force, Web Attack–XSS, Web Attack–SQL Injection, and Benign. Figure 5 illustrates the class-wise distribution of traffic in the dataset, highlighting the degree of imbalance between benign and various attack categories, which is reflective of real-world scenarios. A significant skew toward benign traffic is evident and such imbalance can introduce bias in learning algorithms toward the predominant class, thereby underlining the need for data balancing strategies to ensure adequate representation.

4.1.2 Evaluation of data balancing techniques

Data balancing techniques for severe class imbalance are systematically assessed including oversampling, undersampling, and hybrid approaches, for key metrics of speed, dataset size, memory, and diversity in CIC-IDS 2017 [9] with an emphasis on minority web attacks SQL injection, XSS and Brute Force.

ROS and RUS are excluded due to issues of overfitting and information loss, respectively which are reflected in inadequate web attack recall despite efficiency. Figure 6 illustrates the data balancing comparison among Original, SMOTE, ADASYN, and

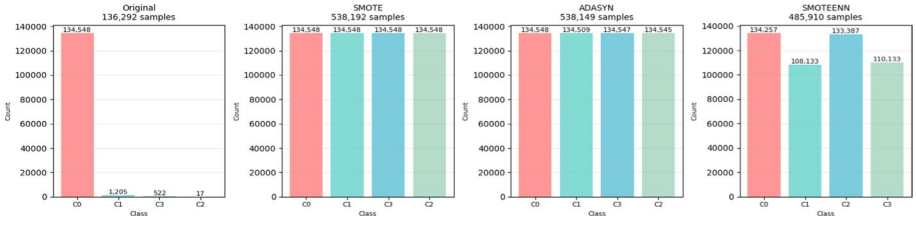


Fig. 6 Data balancing comparison for original, SMOTE, ADASYN and SMOTEEN

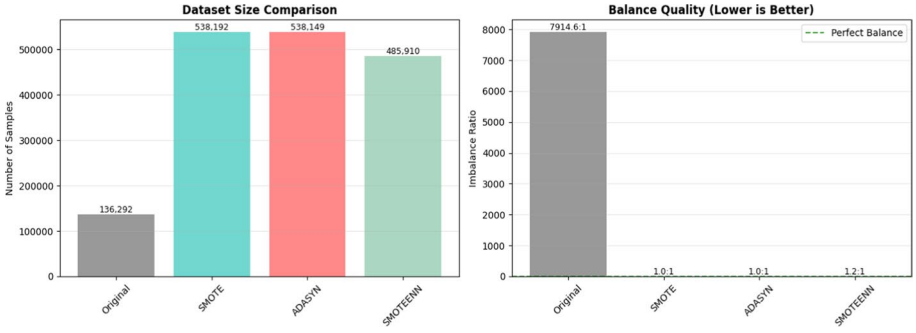


Fig. 7 Dataset size comparison and balance quality

SMOTEENN, emphasizing the pronounced imbalance in the original dataset and the enhancements attained via synthetic data generation.

Figure 7 illustrates the variations in dataset size, highlighting that SMOTE and ADASYN significantly increase sample counts for balance, while SMOTEENN produces a smaller yet refined dataset by eliminating noisy and overlapping samples. Additionally, the balance quality score quantitatively assesses the effectiveness of each method, demonstrating that SMOTE provides a strong baseline, ADASYN enhances representation in complex areas, and SMOTEENN attains the highest balance quality by addressing both class imbalance and data noise.

Further, the results of evaluation of balancing techniques are detailed below.

4.1.3 Random over-sampling (ROS)

In CIC-IDS 2017 [9] for Thursday Web attacks, ROS demonstrates high efficiency ~0.17 s by enlarging the dataset to approximately 672,000 samples, achieving a 3.95-fold increase through minority class duplication to match the majority class size of 168,000, resulting in a memory footprint of around 400 MB and improved minority class representation,however, ROS retains the original minority samples unaltered, preserving the distribution shape but failing to introduce new variability, which may elevate the risk of overfitting despite realizing perfect class balance with limited variability.

4.1.4 Random under-sampling (RUS)

RUS is the fastest observed technique amongst as it condenses the dataset to an extremely small size by downsampling all classes to the count of the rarest class, 84. In [9] RUS reduces the original dataset by diminishing the majority class. While computationally efficient, it discards large amounts of benign traffic, resulting in a high risk of losing informative patterns and degrading model generalization.

Technique	Original Size	Resampled Size	Size Change	Processing Time	Memory (MB)
Random Over-Sampling (ROS)	170,231	672,204	3.95x	0.12s	405.15
Random Under-Sampling (RUS)	170,231	84	0.00x	0.01s	0.05
SMOTE	170,231	672,204	3.95x	0.43s	405.15
SMOTEENN	170,231	671,976	3.95x	394.26s	405.01
ADASYN	170,231	672,191	3.95x	0.91s	405.14

Fig. 8 Comparison of balancing techniques on CIC IDS 2017 [9] dataset

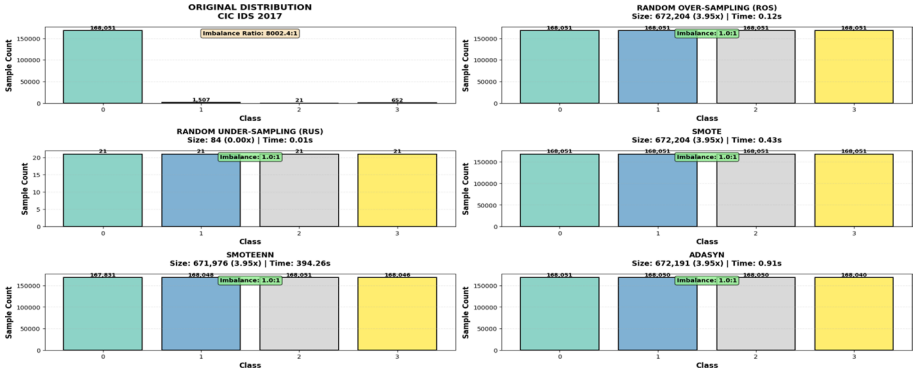


Fig. 9 Visualization for CIC IDS 2017 [9] dataset

4.1.5 SMOTE

SMOTE creates balanced class distributions by interpolating minority samples instead of duplicating them. It operates in approximately 70 s, yielding a dataset size and memory akin to ROS at 672,000 samples. Relative to ROS, SMOTE enhances minority representation through controlled diversity at a moderate computational expense that scales with the dataset's dimensions, though it requires slightly more time for synthetic sample generation via interpolation.

4.1.6 SMOTEENN

SMOTEENN merges SMOTE with Edited Nearest Neighbors (ENN) to eliminate noisy or ambiguous samples post-oversampling. In the CIC-IDS 2017 [9] dataset, it results in a marginally smaller dataset of ~671,976 samples at a significant computational cost ~363 s. Although this hybrid approach enhances data quality by discarding ambiguous samples and minimizing false positives, its extensive processing time renders it less viable for large-scale or time-sensitive contexts, despite similar memory usage..

4.1.7 ADASYN

ADASYN tailors synthetic sample generation towards hard-to-learn minority regions. It operates efficiently in ~0.96 s and produces a dataset size comparable to SMOTE. ADASYN is particularly effective in scenarios where minority classes are highly heterogeneous and decision boundaries are complex. It generates a dataset size similar to both ROS and SMOTE 672,191 samples, while emphasizing hard-to-learn minority regions.

Figure 8 illustrates the Data balancing technique, original size, resampled size, the percent change, processing time taken and memory consumed, evaluating the techniques across these parameters. Figure 9 illustrates evaluation of data balancing across various class labels.

The four-panel visualization in Fig. 10 highlights the trade-offs inherent in various data balancing techniques. The first panel indicates that SMOTEENN is markedly slower, requiring 300+ seconds than alternative methods, making it unsuitable for time-sensitive IDS applications. In contrast, RUS is extremely fast, but incurs significant data loss, while SMOTE and ADASYN present a reasonable balance between computational cost and data integrity. The second and third panels examine dataset size variations, showing that RUS significantly reduces the dataset, while all oversampling methods increase it, impacting memory and storage but remaining manageable with current technology. The final panel evaluates the reduction of imbalance, demonstrating that all techniques successfully transform the original severe imbalance (≈8000:1) into a balanced 1:1 distribution.

Overall, this comparative analysis shows that while every technique corrects class imbalance, SMOTE and ADASYN provide the favorable trade-off between speed, scalability, and data quality. SMOTEENN is impractical for scalability as its excessive runtime offers marginal gains at prohibitive cost, ROS is best suited for rapid prototyping, and RUS is unsuitable due to extreme information loss. Based on this analysis, ADASYN is chosen empirically as it addresses sparse and hard-to-learn minority regions that dominate extreme class imbalance in intrusion datasets. Certain attack categories have as few as 21 samples, and ADASYN generates additional synthetic samples precisely where data density is lowest and provides greater coverage of underrepresented attack manifolds, directly addressing this risk.

4.1.8 Feature engineering

The phase presents the assessed Pearson correlation heatmaps and evaluations of PCA and LDA models.

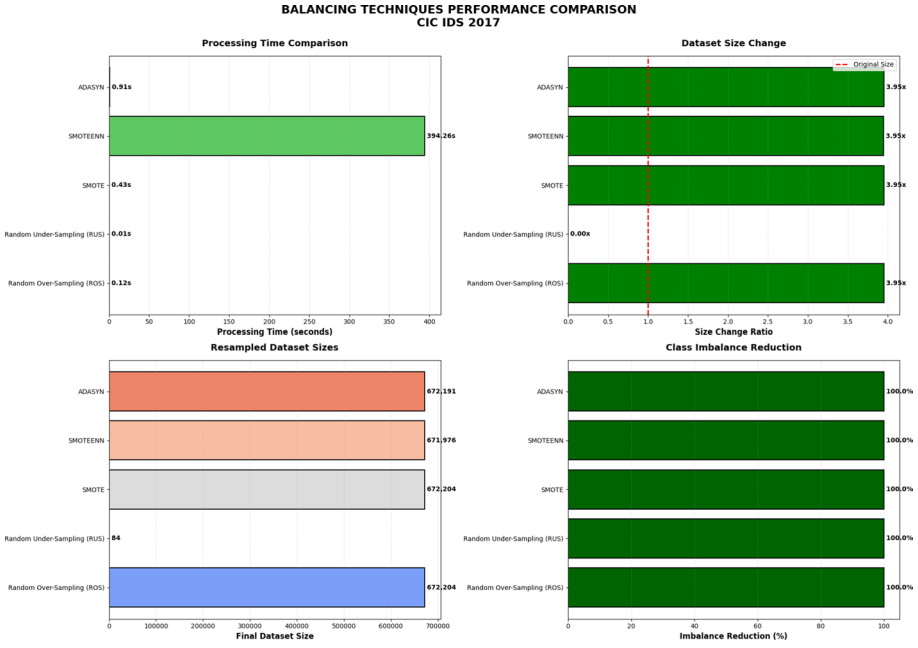


Fig. 10 Performance comparisons for CIC IDS 2017 [9] dataset

4.1.8.1 Pearson correlation matrix evaluation Figure 11 illustrates strong linear correlations in Segment 1 among packet count and length features with a perfect correlation (1.00) between total_backward_packets and total_fwd_packets indicating symmetric bidirectional traffic. Figure 12 indicates significant linear correlations between backward packet length statistics and flow inter-arrival time features in Segment 2. The high correlation coefficients among bwd_packet_length_mean, bwd_packet_length_max, and bwd_packet_length_std (0.83–0.94) suggest a redundancy in representing backward packet size variability. Similarly, the strong correlation between flow_iat_max and flow_iat_std (0.89) demonstrates that extreme inter-arrival times significantly impact overall IAT variability.

Figure 13 examines Segment 3, demonstrating significant temporal interdependencies between forward and backward IAT features, with an almost perfect correlation (0.97) between fwd_iat_min and fwd_iat_mean, suggesting consistent packet timing, and robust correlations (0.89–0.98) between forward and backward IAT statistics indicating synchronized timing behaviors, in addition to a correlation (0.80) between bwd_iat_max and bwd_iat_mean, implying that extreme delays influence average backward IAT values. Figure 14 investigates Segment 4, revealing a perfect correlation (1.00) between bwd_header_length and fwd_header_length, which denotes proportional increases and identical information in header lengths, indicative of symmetric protocol behavior or consistent structures in the analyzed traffic, emphasizing redundancy.

Figure 15 demonstrates significant multicollinearity among packet length statistics, indicating feature redundancy and limited additional discriminatory power. Figure 16 illustrates a strong positive correlation (0.89) between avg_bwd_segment_size and

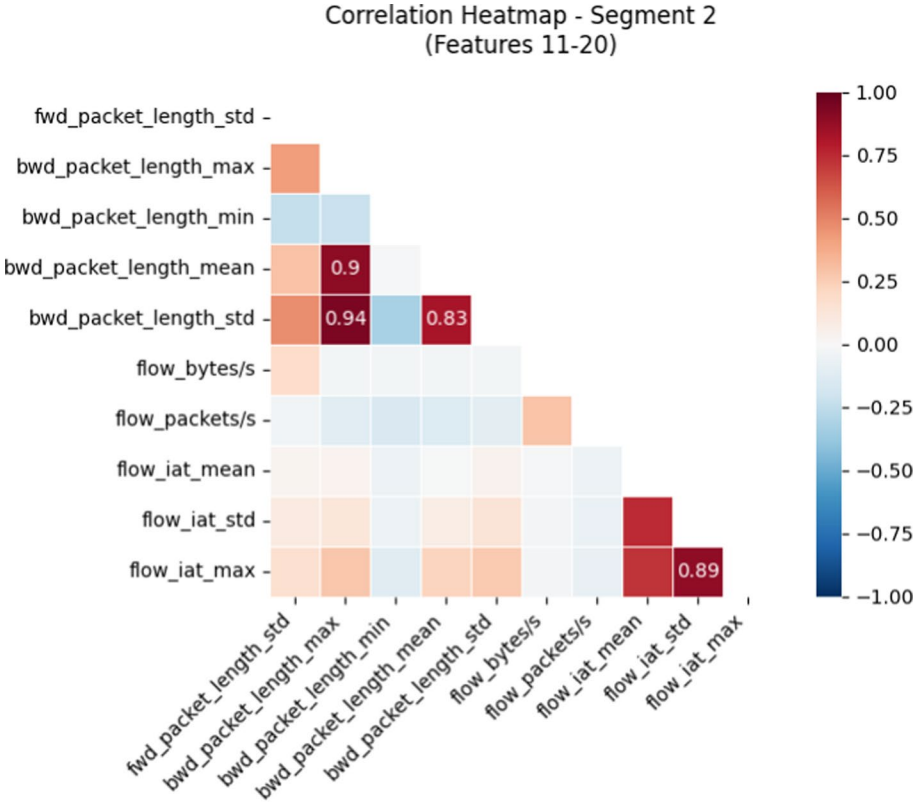


Fig. 11 Correlation heatmap: Segment 1

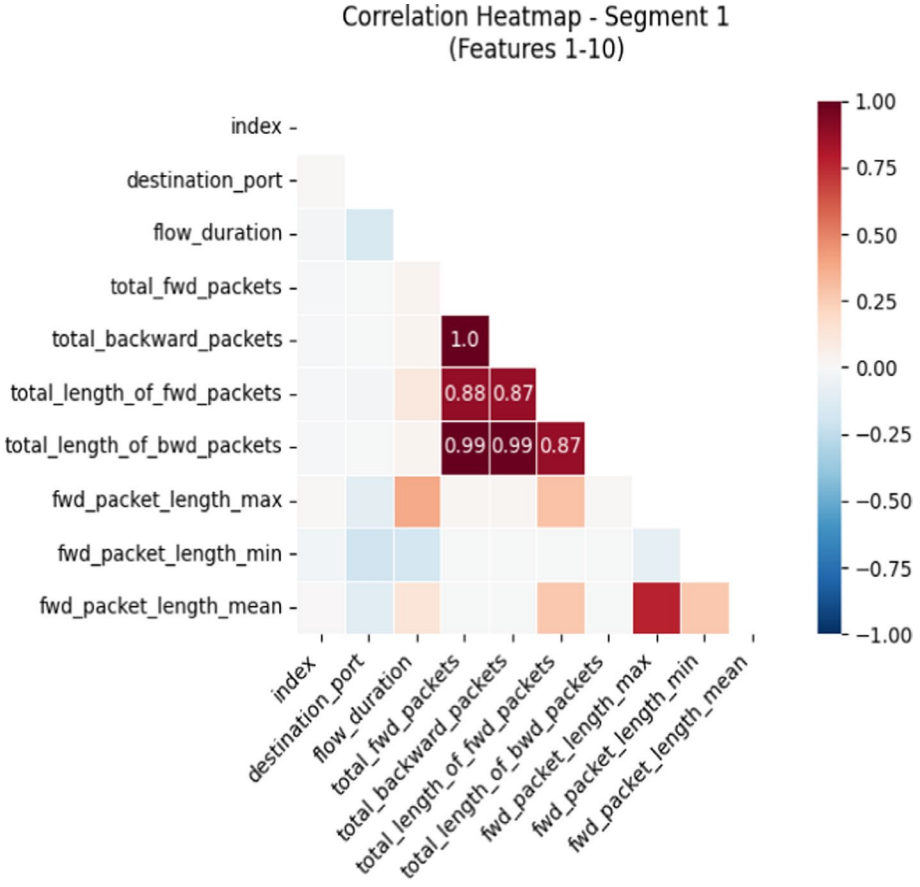


Fig. 12 Correlation heatmap: Segment 2

average_packet_size, suggesting substantial overlap in the information captured by these features regarding traffic structure.

Figure 17 illustrates perfect correlations among subflow-level packet and byte statistics, highlighting significant feature redundancy. The high correlation with act_data_pkt_fwd (up to 1.00) demonstrates a close representation of subflow packet activity. Consequently, these findings suggest substantial collinearity and interchangeability among subflow-level features. The correlation analysis for Segment 8 in Fig. 18 indicates robust linear relationships among subflow-level features, with high correlations reflecting proportional scaling of packet and byte counts across subflows, while act_data_pkt_fwd effectively mirrors subflow activity information.

For CIC-IDS 2017 [9] Thursday Web Attacks, features exhibiting high variance indicate that attack traffic is characterized by longer durations and irregular inter-arrival timings, establishing duration and throughput as critical discriminators for anomaly detection.

4.1.8.2 Comparison of PCA and LDA for feature extraction Figure 19 depicts that Principal Component Analysis (PCA) achieves significant speed enhancements; however, this comes at the cost of considerable degradation in accuracy and F1-score, rendering it inadequate for dependable intrusion detection. In contrast, Linear Discriminant Analysis (LDA) maintains performance that is nearly consistent with baseline metrics ≈98% accuracy, and ≈98.6% F1 while incurring only minimal computational overhead, thereby

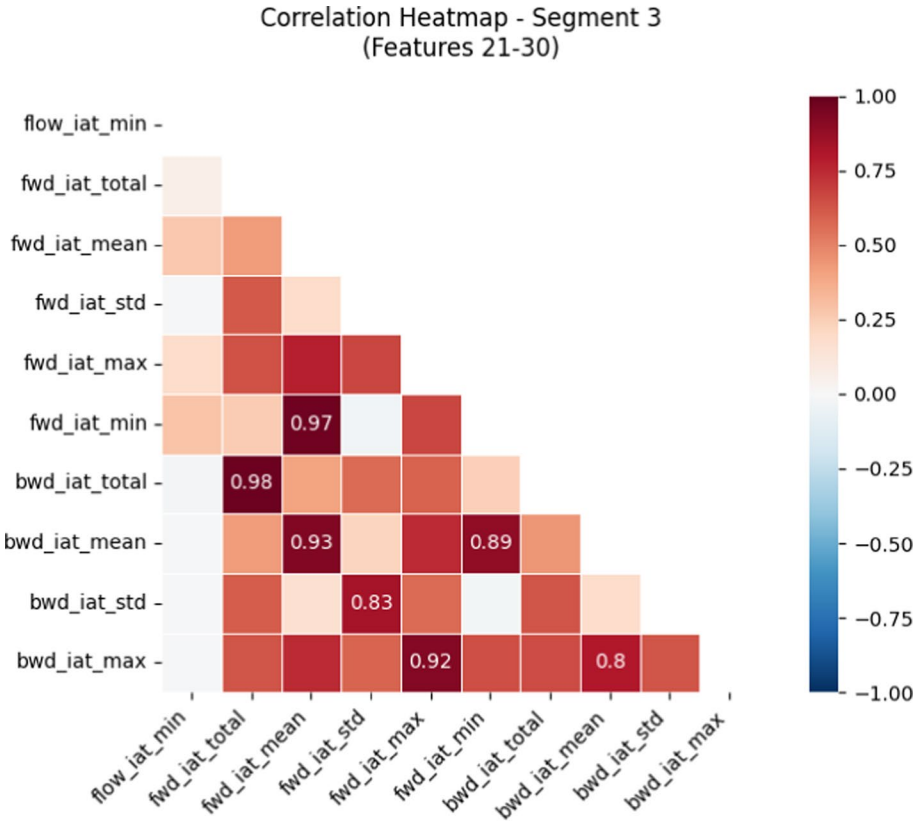


Fig. 13 Correlation heatmap: Segment 3

offering the most favorable balance between operational efficiency and detection quality among all examined models.

Models trained on the original 79-dimensional feature set achieve the highest classification accuracy ≈99% across CNN, LSTM, and DNN architectures. Minority class recall is strong, indicating that the full feature space retains detailed information necessary to detect rare attack patterns. However, this performance comes at the cost of high computational complexity, increased memory consumption, and slower inference, limiting the feasibility of deploying in real-time, resource-constrained environments.

PCA reduces the feature space to three unsupervised components, achieving a dimensionality reduction of 96.2%. While this results in notable speedups, it introduces significant performance degradation across all models. Accuracy drops range from 9% to 15%, and confusion matrices show extensive off-diagonal dispersion, indicating poor class discrimination. It fails to preserve minority class characteristics, as it prioritizes variance rather than class separability, leading to increased misclassification, making it unsuitable for security-critical applications.

LDA also reduces the feature space to three supervised components (n – 1 classes) while explicitly maximizing between-class separability. Although LDA introduces an accuracy reduction of approximately 1% compared to the original feature space, it significantly outperforms PCA across all architectures.

Confusion matrix in Fig. 20 analysis reveals strong diagonal dominance for LDA-based models, indicating clear class boundaries and minimal inter-class confusion. Despite the

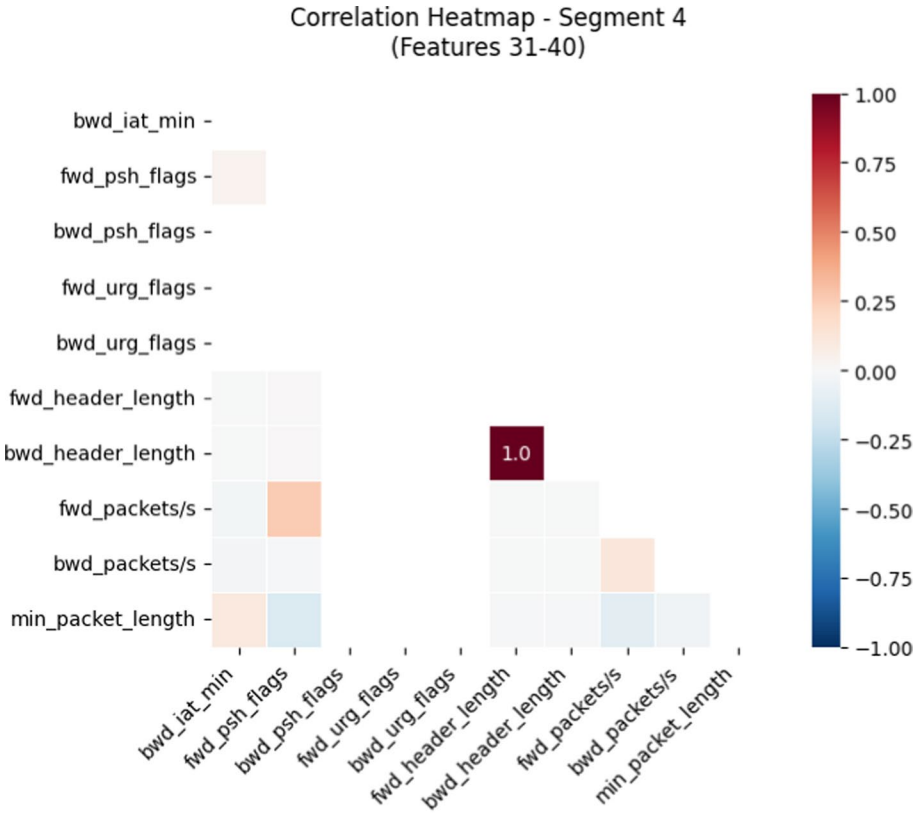


Fig. 14 Correlation heatmap: Segment 4

aggressive dimensionality reduction, LDA maintains well-structured decision regions and achieves the best F1-scores among reduced-feature approaches.

The comparative results demonstrate that LDA offers the optimal trade-off between performance and complexity:

Accuracy: Maintains ≈98% accuracy with acceptable ~1% degradation from the original feature space
Class separability: Superior confusion matrix structure compared to PCA
Efficiency: 96.2% feature reduction enables faster inference and lower memory usage
Stability: Consistent performance across CNN, LSTM, and DNN architectures
Practicality: Well-suited for real-time and large-scale intrusion detection deployments

While the original feature representation yields the highest raw accuracy, its computational demands limit practical deployment. PCA, although efficient, fails to preserve class-discriminative information and significantly degrades performance. LDA emerges as the most balanced and deployment-ready solution, achieving strong classification performance with minimal accuracy loss and substantial efficiency gains. Consequently, LDA is the preferred dimensionality reduction technique for scalable and real-time intrusion detection systems.

4.1.9 Evaluation of classification models

The study evaluates optimizers such as Adam, RMSprop, and SGD to determine the best update rule for models. Hyperparameters, including learning rate and batch size, are optimized using manual experimentation and grid search based on validation metrics.

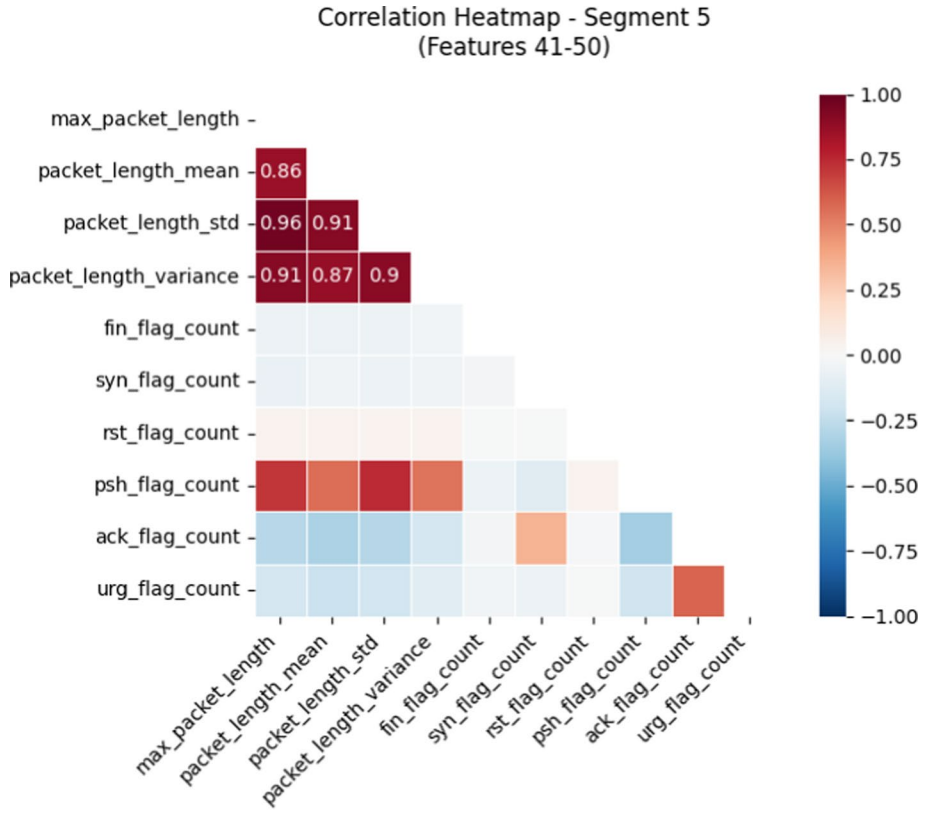


Fig. 15 Correlation heatmap: Segment 5

These optimizations are guided by loss and accuracy metrics to ensure effective generalization. Regularization methods like dropout layers and early stopping are utilized to strengthen model resilience and reduce overfitting.

To optimize the deep learning architecture, a defined hyperparameter search space is created, exemplified for the 1D CNN in Fig. 21, including various architectural, optimization, and training parameters. This search space offers around 240,000 potential configurations, and manual strategies alongside pruning are employed to efficiently navigate these extensive search spaces, as pure computational reliance is resource-intensive.

To analyze the model's learning behavior and convergence, training and validation curves for accuracy and loss versus epochs are plotted. The confusion matrix visualizes prediction distribution across classes, facilitating an intuitive understanding of misclassification, which may not be fully captured by accuracy metrics alone. Evaluation strategy and its relevance is depicted in Table 4.

Various hyperparameter combinations are evaluated. The same hyperparameter set is used across all models. The CNN exhibits consistent performance across all scenarios tested. Three ablation studies focus on different optimization algorithms: SGD, Adam, and RMSProp are detailed. The CNN excels in accuracy and recall when using SGD and Adam within the preprocessed pipeline. In contrast, the DNN slightly outperforms in accuracy with RMSProp, but at a cost of increased execution time.

Case 1: An ablation study of 1000 epochs with a batch size of 512, featuring 64 input layer units and 128 dense layer units, utilizes a dropout rate of 0.2 and a learning rate of 0.01. The loss function employed is Sparse Categorical Cross entropy. The results demonstrate that CNN, LSTM, and DNN exhibit strong performance on the

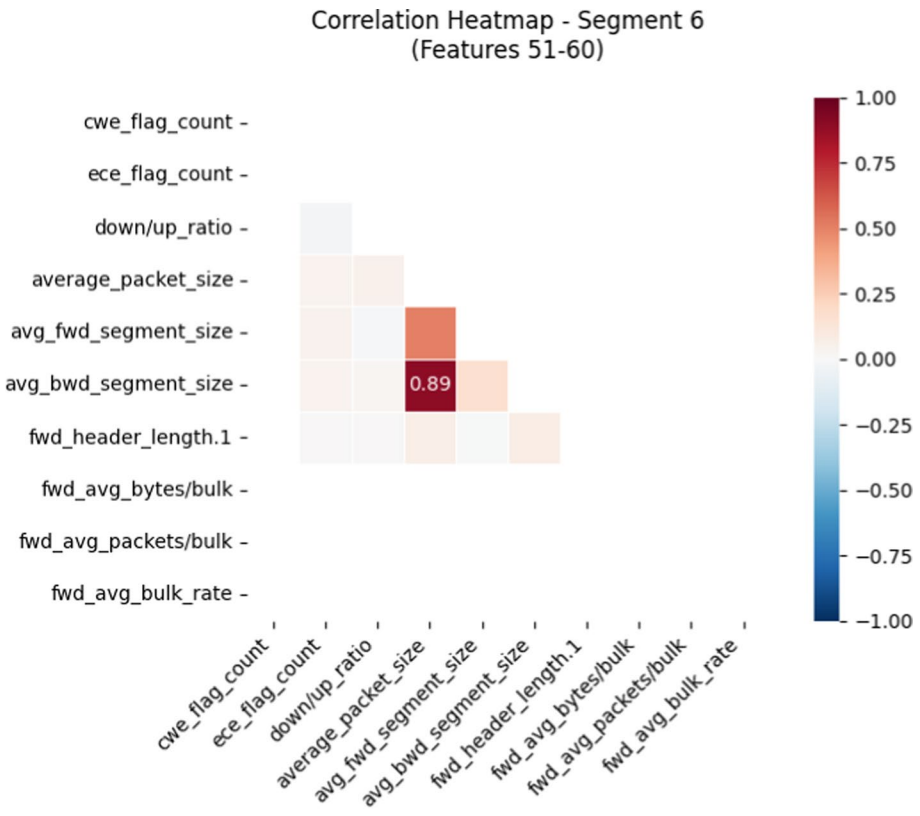


Fig. 16 Correlation heatmap: Segment 6

ADASYN-balanced and LDA-transformed feature space, with accuracies exceeding 95%. Among these, CNN achieves the highest accuracy at 95.3%, marginally better than LSTM with 95.1% and DNN with 95.0%. The minor performance difference still positions CNN as the superior model for this application. In terms of training efficiency, CNN converges fastest, requiring about 13 minutes, followed by DNN at 19 minutes and LSTM at 31 minutes. The extended duration of LSTM training is due to its recurrent architecture and gate-based computations, which are computationally intensive despite dealing with static features. This illustrates the trade-off between model complexity and computational efficiency.

Figure 22 illustrates the comparative analysis, assessing CNN attains the highest accuracy and recall at 95.39%, LSTM achieves the optimal precision of 99.39%, and DNN yields superior F1-score of 97.19%. The elevated recall associated with CNN signifies an exceptionally low false-positive rate, which substantiates effective identification of attack instances.

The negligible performance gap among CNN, LSTM, and DNN suggests that the greater contributor to detection effectiveness is the previous pipeline. The use of ADASYN-based oversampling effectively mitigates class imbalance, while LDA-based feature extraction successfully reduces dimensionality and enhances class separability. Together, these steps enable all models to learn meaningful and discriminative representations.

Precision plots in Fig. 23 are ≈99.3%, indicating a minimal false-positive rate, while recall values from 95.1 to 95.4% demonstrating the models' efficacy in detecting most attack instances. F1-scores are near 97%, indicating a strong balance between precision

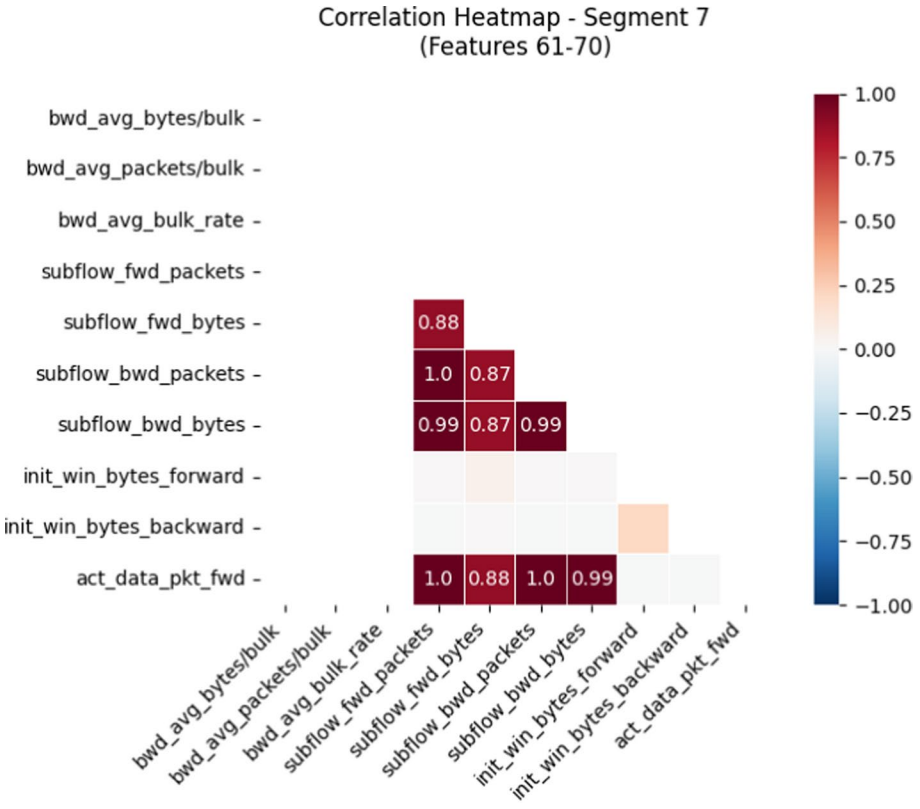


Fig. 17 Correlation heatmap: Segment 7

and recall across various architectural configurations. A comparative analysis of training times is depicted in Fig. 24.

Case 2: The illustration in Fig. 25 presents a comparative performance analysis of CNN, LSTM, and DNN architectures trained using the RMSprop optimizer under identical uniform and evaluation settings. Among the analyzed models, the DNN attains the highest accuracy of 96.27% and F1-score of 97.67%, with the CNN closely following at 96.12%, accuracy and F1 score of 97.58%. The LSTM records marginally inferior yet competitive performance recording an accuracy of 95.47% and an F1-score of 0.9726. The adaptive learning rate mechanism of RMSprop appears to be particularly efficacious for the DNN architecture.

The visualization in Fig. 26 depicts precision–recall scatter plot further highlights the high-quality classification behavior of all models, with precision values exceeding 99.4% and recall values above 95.0%.

Case 3: Fig. 27 delineates the specifics of the performance comparison, highlighting the efficacy of three distinct neural network architectures that have been subjected to training under a framework that ensures methodological rigor through the application of uniform hyperparameters, an identical optimizer configuration, a standardized preprocessing strategy, and a consistent evaluation protocol across all models, utilizing the Adam optimizer. The Convolutional Neural Network (CNN) attains the highest accuracy of 96.26% and an F1-score of 97.67%, thereby marginally surpassing the performance metrics of both LSTM and DNN architectures.

Figure 28 represents the multi-panel visualization and provides a holistic comparison. In lieu of concentrating exclusively on accuracy, the analysis also assesses architectural

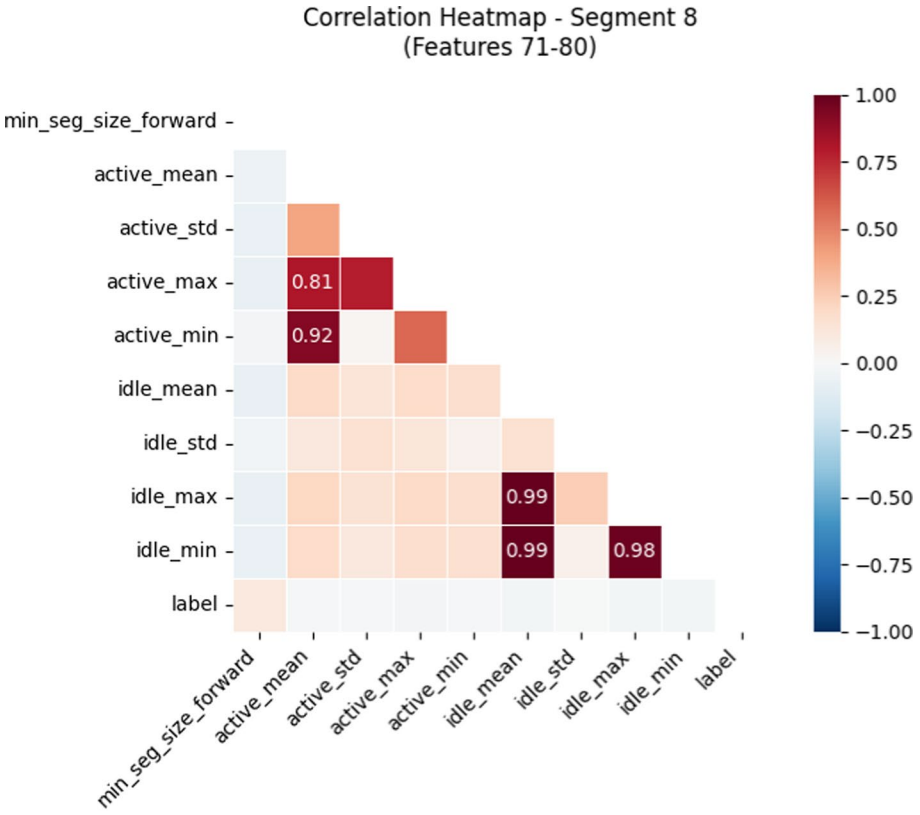


Fig. 18 Correlation heatmap: Segment 8

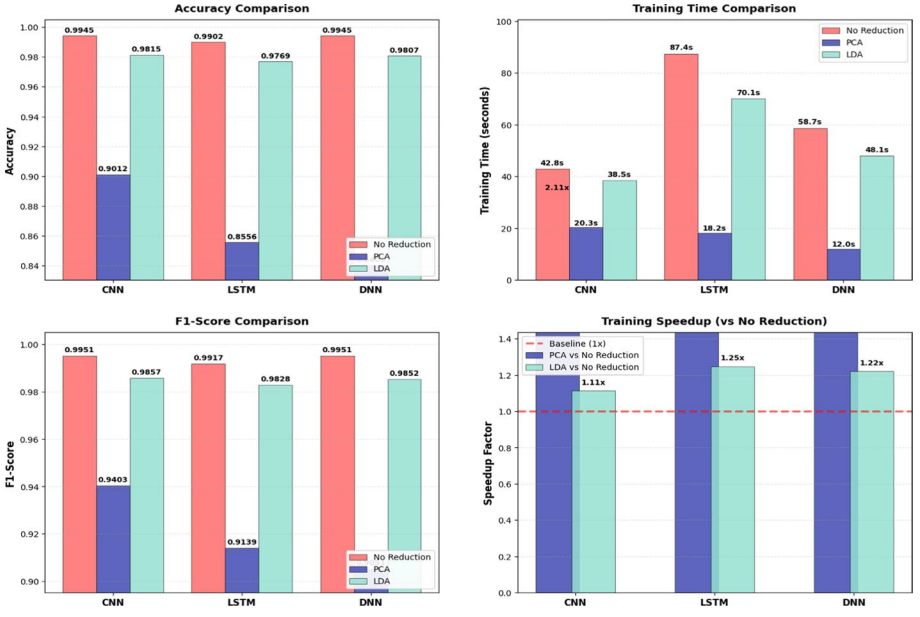


Fig. 19 Performance visualization for original vs. PCA vs. LDA

complexity, computational efficiency, training behavior, and the appropriateness of each model for large-scale intrusion detection applications. The complexity and parameter analysis shows that the LSTM is the most complex architecture, followed by the DNN, whereas the CNN is characterized as the simplest model possessing nearly half

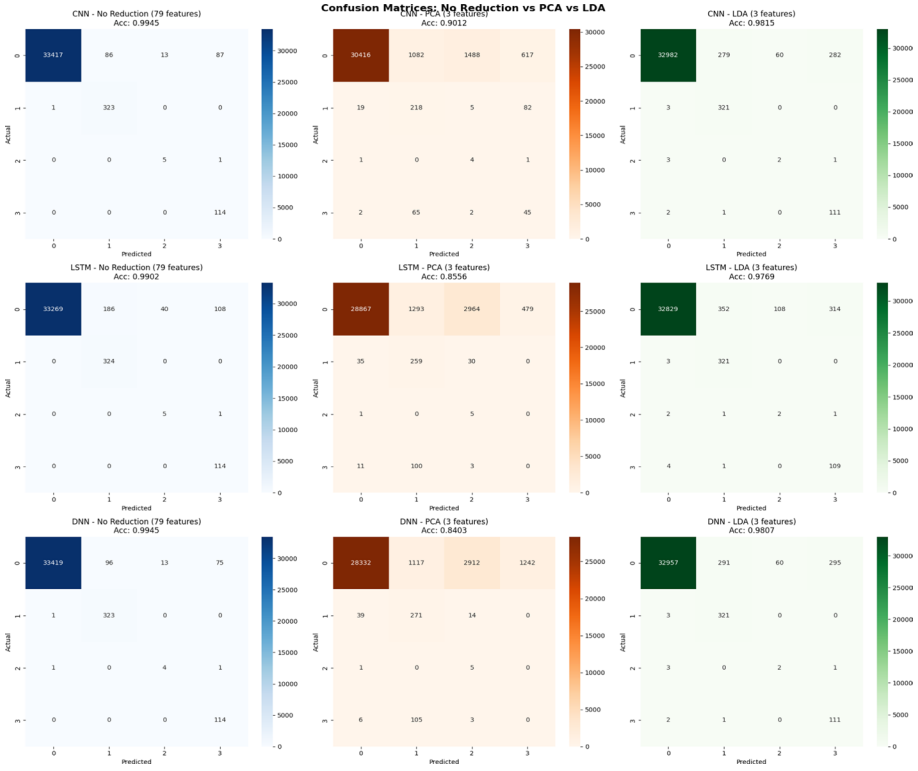


Fig. 20 Confusion matrix evaluation for original vs. PCA vs. LDA based reduction

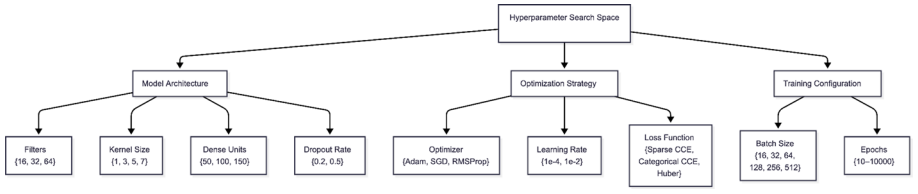


Fig. 21 Hyperparameter search space for 1D CNN

Table 4 Evaluation strategy and its importance

Evaluation strategy	Importance
Classification accuracy	Correct predictions across all classes
Precision, recall, F-1 score	Per class computation to ass the detection of quality and balance
Confusion matrix	Tabular representation of model performance across attack types
Training and validation graphs	Accuracy and Loss plots used for monitoring learning and progression stability
Comparative analysis	Performance benchmarking across different architectures and optimization strategies

the number of parameters in comparison to its counterparts. Despite this reduced complexity, the CNN achieves superior classification performance, highlighting its efficiency in learning discriminative patterns from the LDA-transformed feature space. This effectiveness is further substantiated in the context of training time, wherein the CNN has the fastest convergence, the DNN incurs moderate training costs, and the LSTM necessitates significantly longer durations due to its inherent sequential processing characteristics.

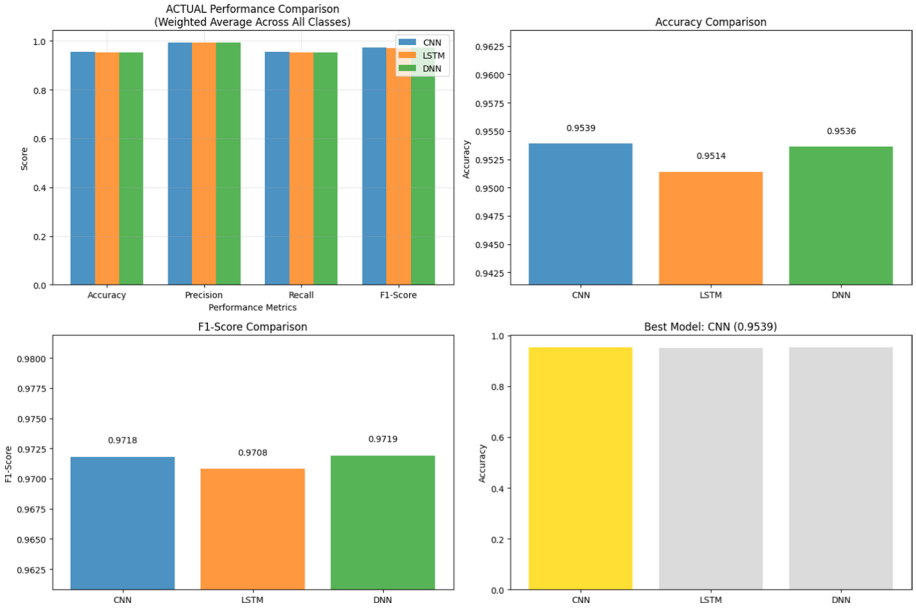


Fig. 22 Evaluation of deep learning models for SGD optimizer

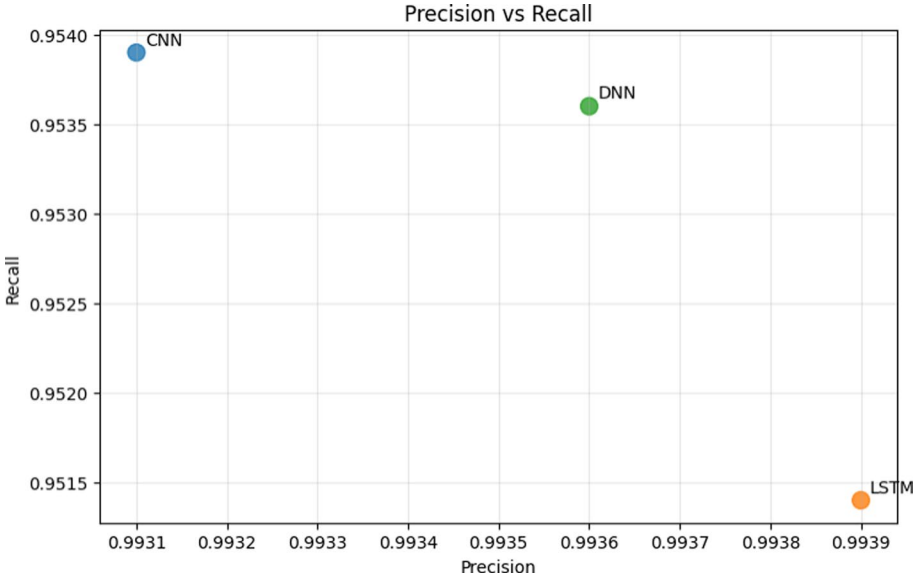


Fig. 23 Precision vs. recall plot

4.1.9.1 Cross-validation analysis of CNN performance The six-panel visualization in Fig. 29 presents a comprehensive cross-validation (CV) analysis of the CNN model, employing a tenfold cross-validation methodology to evaluate performance stability, robustness, and statistical reliability.

Panel 1 visualizes CNN Performance across 10 folds illustrate the variability of accuracy, precision, recall, and F1-score across all folds. All performance metrics consistently maintain elevated levels, ranging between 98.5 and 99.5%.This suggests that the CNN model exhibits a high degree of stability and generalization capabilities.

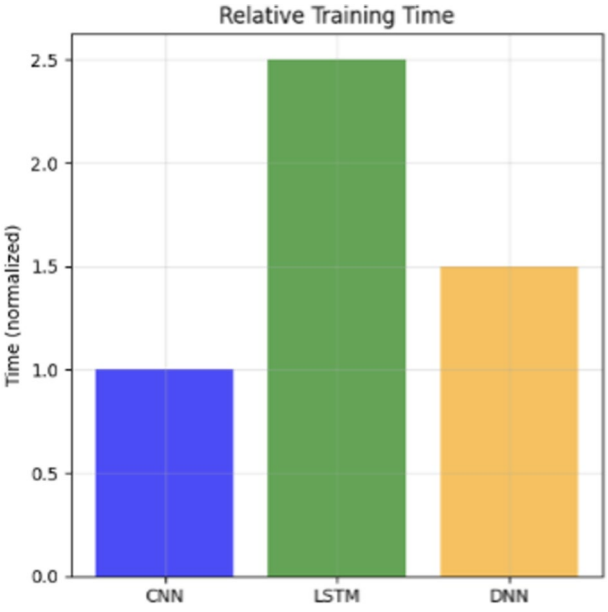


Fig. 24 Relative training time comparison

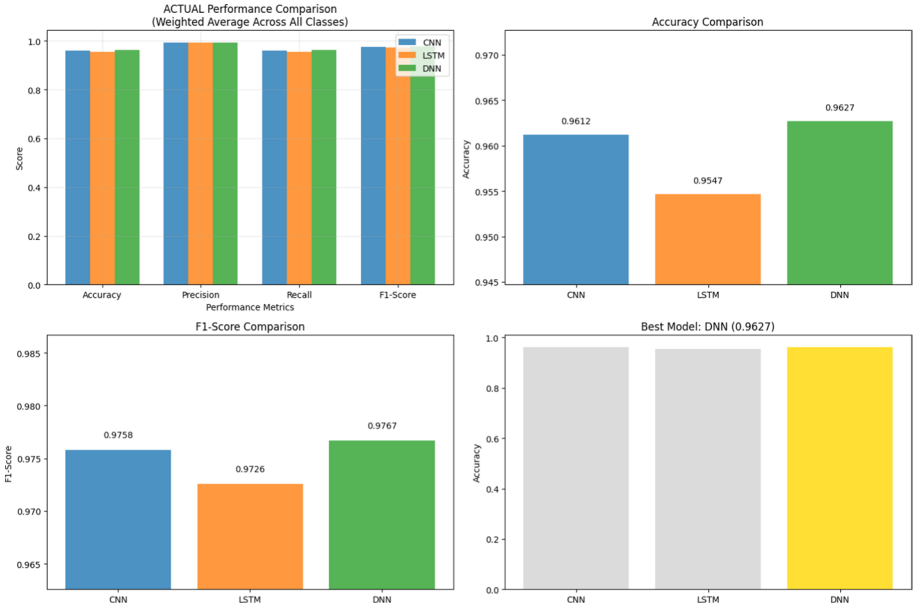


Fig. 25 Evaluation of deep learning models with RmsProp optimizer

Panel 2 illustrates the statistical distribution of each metric across the folds. The limited interquartile ranges and the absence of significant outliers corroborate the findings of low variance and robust performance.

Panel 3 depicts the distribution of accuracy values across folds, with an average accuracy of 98.6%, confined within a narrow range of 98.6–98.8%, and the mean is highlighted by a dashed red line.

Panel 4 delineates the computational cost per fold, with training durations ranging from approximately 7 to 12 min. Minor variations are expected due to convergence behavior and system load.

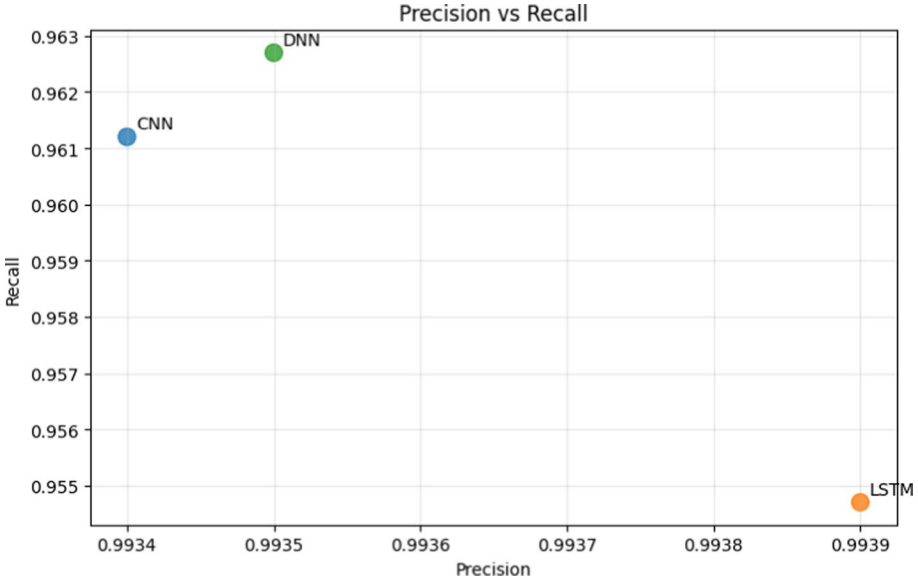


Fig. 26 Precision vs. recall plot

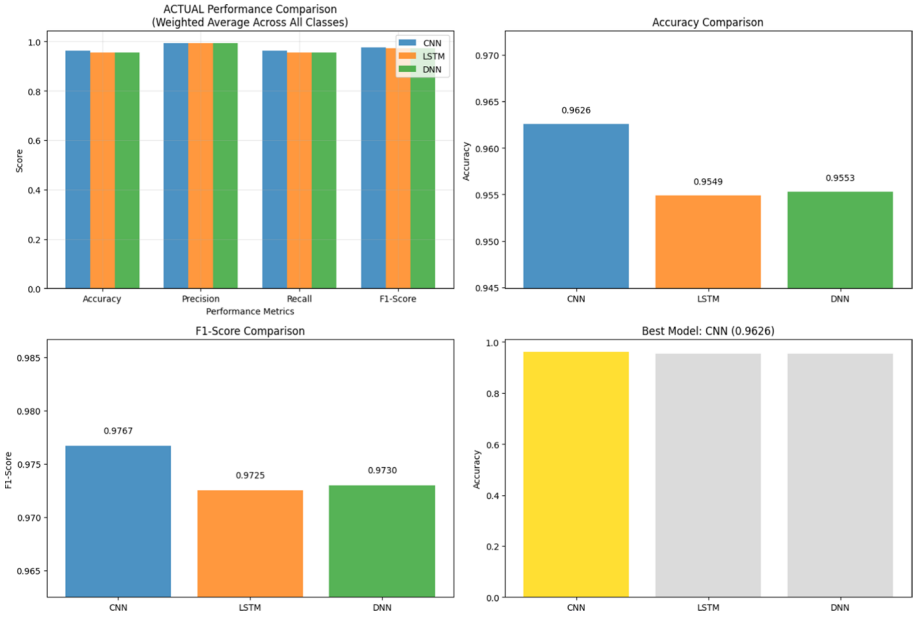


Fig. 27 Evaluation of deep learning model with Adam optimizer

Panel 5 shows mean values of Accuracy, Precision, Recall, and F1-score accompanied by 95% confidence intervals. All metrics attain values nearing 99% with exceedingly narrow confidence bounds (approximately ±0.001), reflecting high statistical confidence and reproducibility of CNN's performance.

Panel 6 compares the original 80–20 holdout accuracy of 96.26% with the cross-validation mean accuracy of 98.66%. While cross-validation entails a higher computational cost relative to a singular holdout evaluation, the resulting robustness and reliability substantiate its use. Collectively, this cross-validation investigation reveals that the CNN model consistently achieves accuracy exceeding 98.5% with minimal variance, thereby offering a statistically validated and highly reliable performance benchmark. These

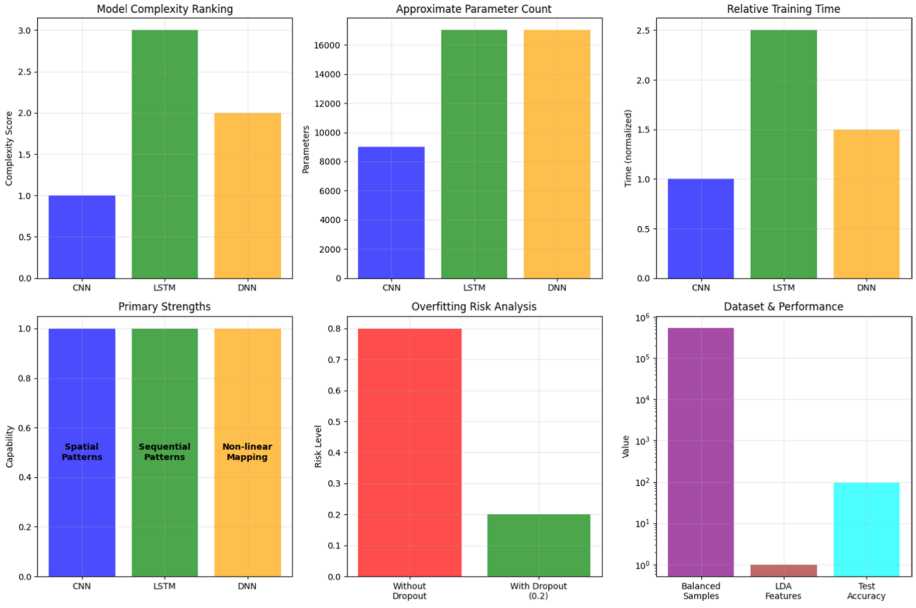


Fig. 28 Model complexity and performance analysis for deep learning models

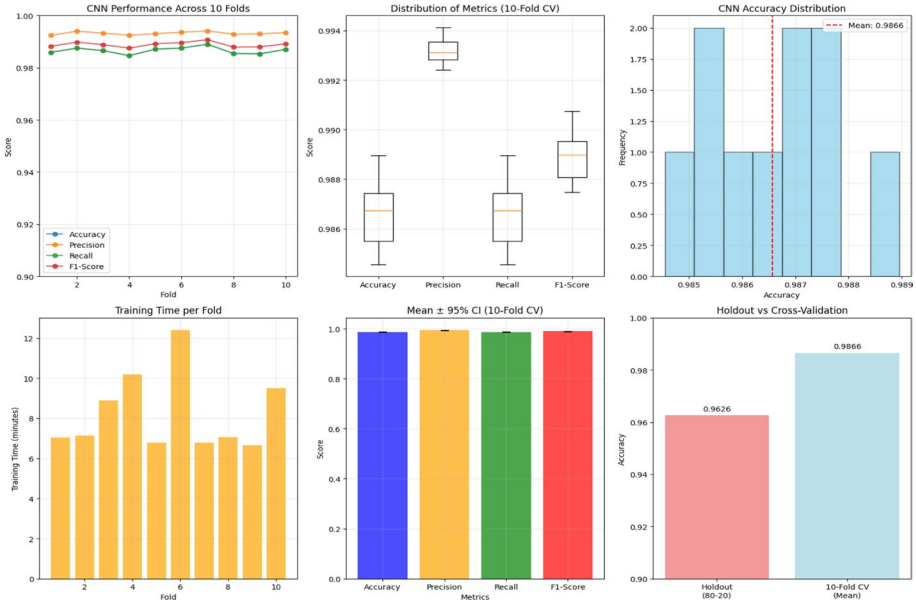


Fig. 29 Cross validation for CNN model

findings establish a strong, reproducible baseline for subsequent comparison with LSTM and DNN architectures, significantly enhancing the scientific rigor of the evaluation.

5 Results and performance analysis

This section presents and analyzes the experimental results of the proposed approach.

5.1 Performance comparison

Utilizing an ADASYN-based preprocessing framework alongside CNN model calibrated with optimized hyperparameters (64 filters, kernel size = 1, 64 dense units, dropout

rate = 0.2, Adam optimizer with a learning rate of 0.001, batch size of 1024, and a total of 10,000 epochs), the model attained a superior test accuracy of 99.79%, a low validation loss of 0.118, and markedly reduced execution time, thereby positioning it as the most suitable candidate for real-time web attack detection.

CNN exhibits superior performance in the classification of prominent attack vectors, achieving flawless detection of benign traffic and robust F1-scores for Brute Force (0.94) and SQL Injection (0.90), while its efficacy on infrequent classes, such a XSS give F1 score of 0.40.

DNN, characterized by a multiple densely interconnected layers utilizing Rectified Linear Unit (ReLU) activations, demonstrate training accuracy of 99.88% yet encountered a decline in validation accuracy of 90.26% alongside an increase in validation loss of 2.24%, thus signifying a tendency towards overfitting and augmented computational demands. Dropout regularization rate of 0.2 is employed subsequent to each dense layer. The model is optimized using the Adam algorithm with a learning rate of 0.01, a batch size of 2048, and a total of 5000 epochs, thereby achieving an equilibrium between convergence velocity and training stability. The DNN model demonstrates high accuracy for the Benign and Web Attack Brute Force classes, with F1 scores of 1.0 and 0.90, respectively. While it maintains a strong F1-score of 0.82 for SQL Injection, its performance on the *XSS* class remained low (0.31), impacted by the limited sample size.

The Long Short-Term Memory (LSTM) model comprises of 256 neurons, with ReLU activation function undergoes training utilizing the adam optimization algorithm with a learning rate of 0.01, a batch size of 1024, and extends over 2000 epochs, attained a test accuracy of 98.07%; however, it recorded the lowest validation accuracy of 79.02% and the most protracted execution time, indicative of diminished generalization and efficiency. The LSTM model achieves perfect classification for the Benign class but showed high recall with low precision for all attack types. It attains an F1-score of 0.70 for Brute Force and 0.50 for SQL Injection, while XSS detection suffered a low F1-score of 0.11, indicating a tendency to overpredict minority classes with limited precision.

Table 5 outlines the models compared to different parameters, including training accuracy and loss, test accuracy and loss, validation accuracy, and loss.

CNN emerged as the most efficient and generalizable model, offering the most advantageous trade-off between accuracy, stability, and computational efficiency for multiclass web attack detection (Table 6).

5.2 Training and convergence behavior

The convergence behavior of the CNN, DNN, and LSTM models is monitored through training and validation of accuracy and loss curves. Despite all three models achieving high training accuracy, their generalization performance varied greatly. The CNN architecture offers a robust generalization capability as it reveals a smooth and consistent convergence with a minimal discrepancy between training and validation outcomes.

Table 5 Performance evaluation for CNN, LSTM and DNN

Model	Training		Test		Validation	
	Accuracy	Loss	Accuracy	Loss	Accuracy	Loss
CNN	99.87	0.0175	99.79	0.0175	97.31	0.118
DNN	99.88	0.0070	99.62	0.899	90.26	2.246
LSTM	91.37	0.1334	98.07	0.133	79.02	0.0525

Table 6 Classification matrix for CNN, DNN and LSTM

Model	Precision				Recall				F1-score				Support			
	0	1	2	3	0	1	2	3	0	1	2	3	0	1	2	3
CNN	1.00	0.89	0.50	0.85	1.00	0.99	0.33	0.95	1.00	0.94	0.40	0.90	33,603	324	6	114
DNN	1.00	0.82	0.29	0.72	1.00	1.00	0.33	0.96	1.00	0.90	0.31	0.82	33,603	324	6	114
LSTM	1.00	0.55	0.06	0.42	0.99	0.98	0.83	0.95	0.99	0.70	0.11	0.58	33,603	324	6	114

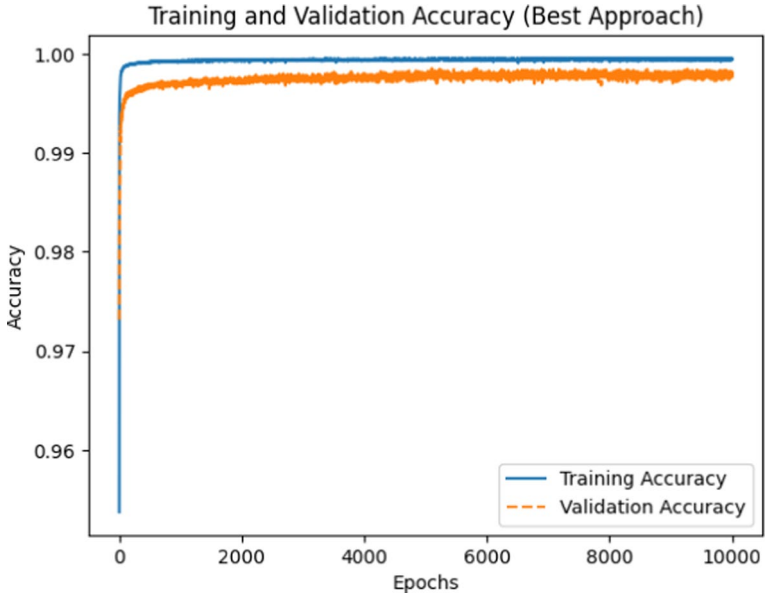


Fig. 30 Training and validation accuracy (CNN)

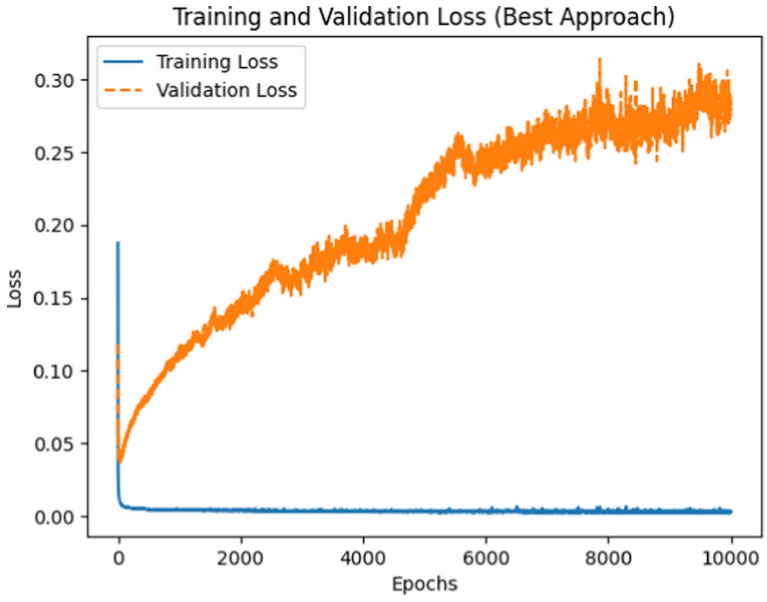


Fig. 31 Training and validation loss (CNN)

The DNN achieves the highest training accuracy; however it demonstrates indications of overfitting, as evidenced by its diminished validation accuracy and increased validation loss in subsequent epochs. In contrast, the LSTM model undergoes a more gradual convergence, higher training time and shows unstable validation performance. These observations highlight the preeminence of CNN model for effective and scalable web attack detection in this experimental setup. Figures 30, 32, and 34 illustrate training and validation accuracy, while Figs. 31, 33, and 35 depict training and validation of loss across epochs for the CNN, DNN, and LSTM models, respectively.

Figures 36, 37 and 38 illustrates the confusion matrices for CNN, LSTM, and DNN classifiers utilizing the ADASYN–LDA preprocessing method. In these matrices, rows



Fig. 32 Training and validation accuracy (DNN)

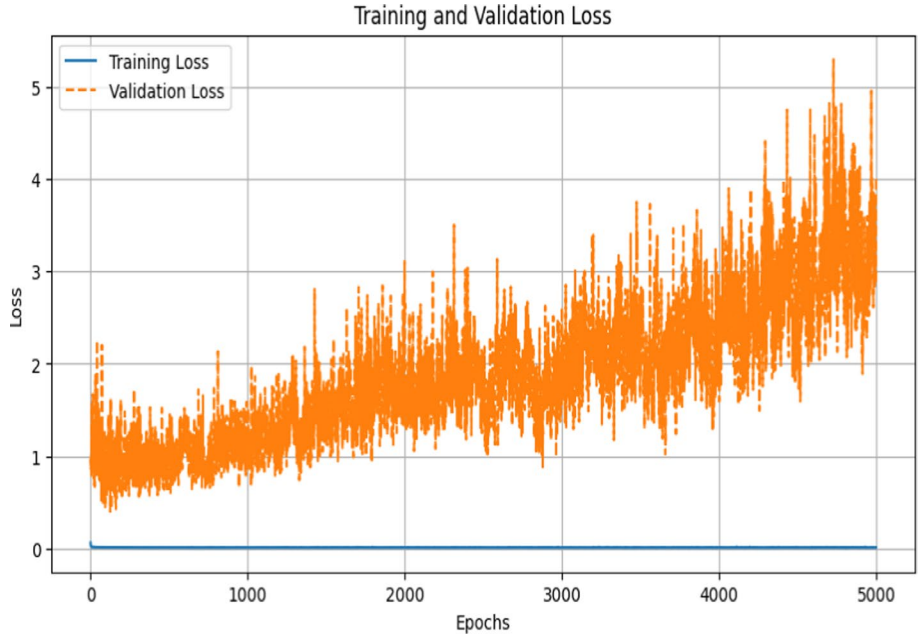


Fig. 33 Training and validation loss (DNN)

indicate true class labels, columns signify predicted labels, and diagonal elements represent correct classifications, while off-diagonal elements denote misclassifications. The majority class, benign traffic (Class 0), is accurately detected by all models, with CNN and LSTM exhibiting the lowest false-positive rates, thus being advantageous for reducing unnecessary security alerts.

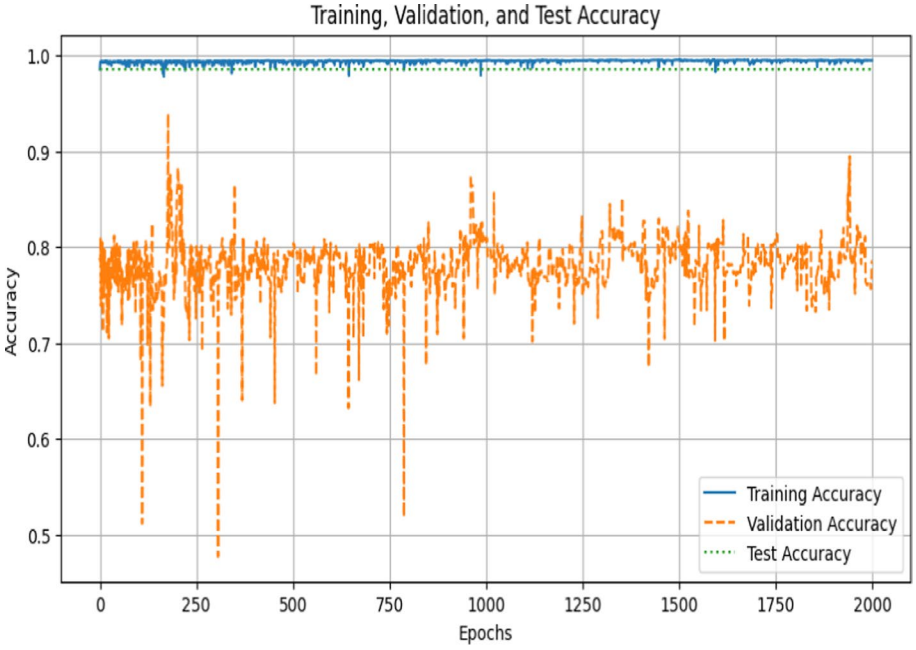


Fig. 34 Training and validation accuracy (LSTM)

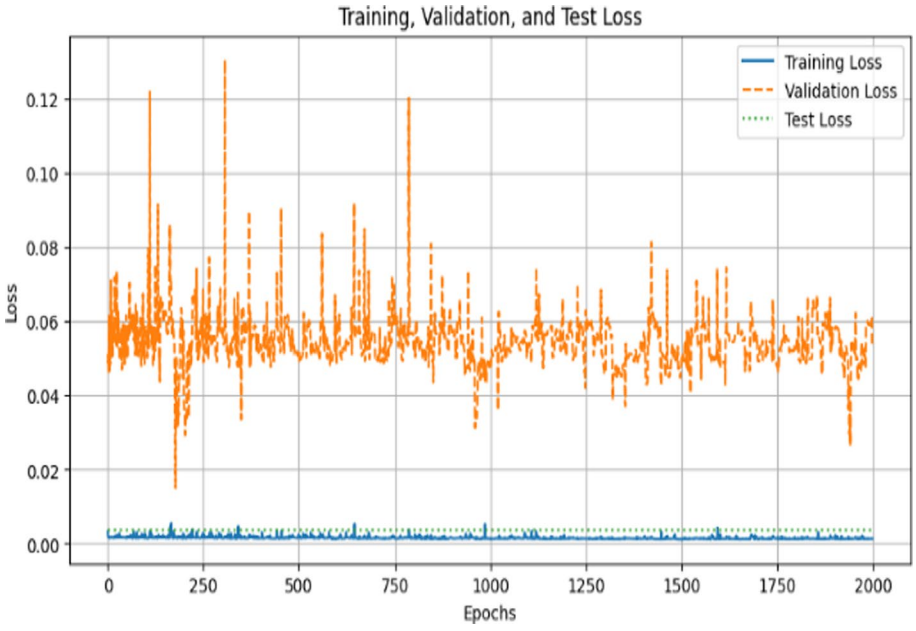


Fig. 35 Training and validation loss (LSTM)

For moderately represented attack classes (Class 1 and Class 3), CNN demonstrates competitive effectiveness. The DNN shows enhanced recall for Class 3. Conversely, CNN achieves a favorable equilibrium between benign traffic identification and attack detection, resulting in the highest overall accuracy and minimal benign-to-attack misclassifications. All models encounter challenges in detecting the rarest class (Class 2).

In conclusion, the confusion matrices reveal a distinct trade-off between overall accuracy and sensitivity to rare attacks. While the DNN enhances detection of specific minority attacks, the CNN provides the most balanced and practically viable

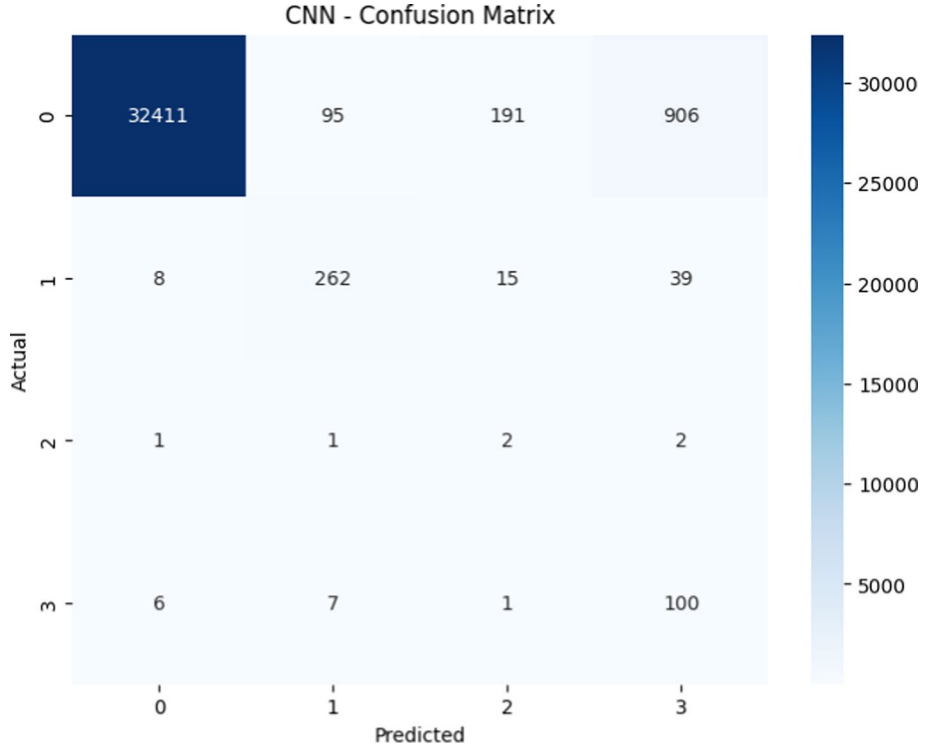


Fig. 36 Confusion matrix for CNN

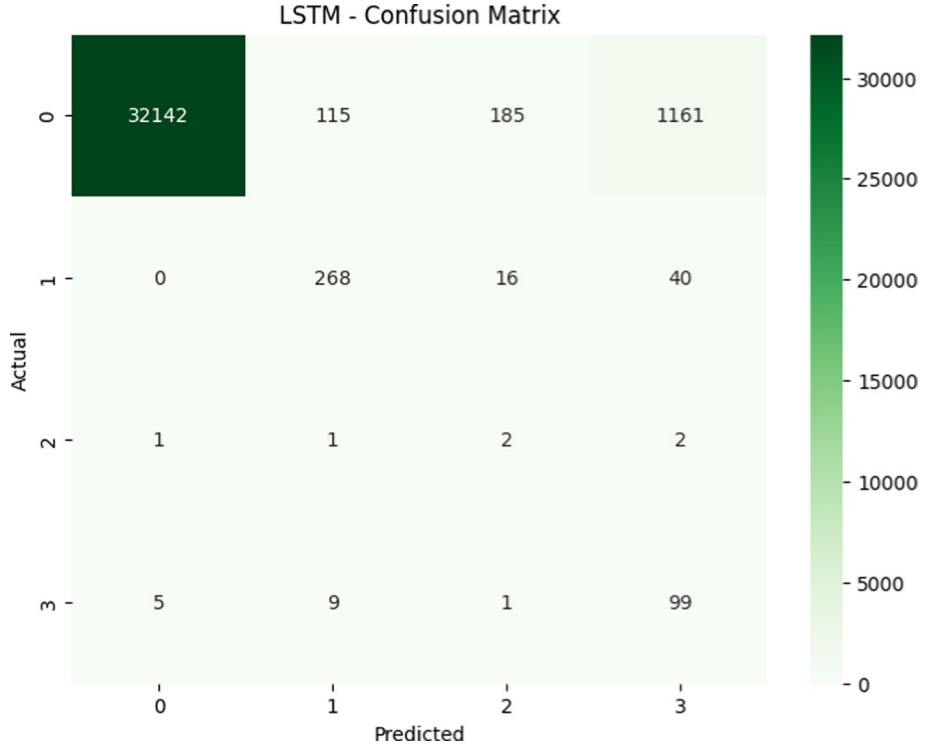


Fig. 37 Confusion matrix for LSTM

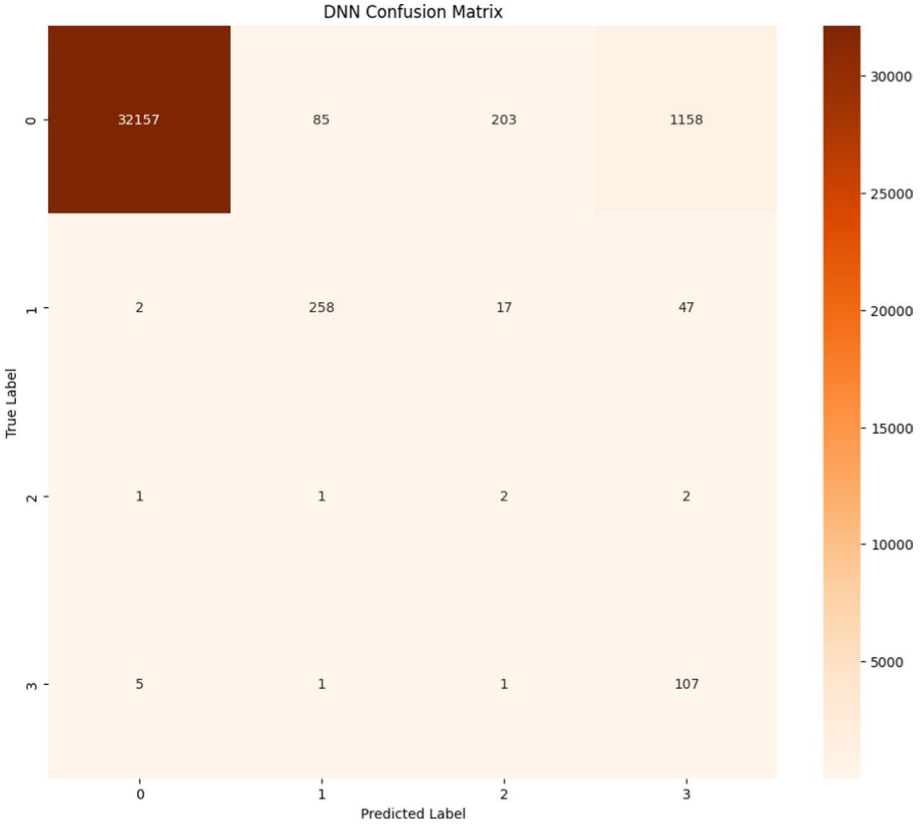


Fig. 38 Confusion matrix for DNN

performance, merging high accuracy in benign traffic, minimized false positives, and robust detection of prevalent attack classes. This analysis substantiates CNN as the most dependable option for implementation in real-world intrusion detection systems, where consistency and low false-alarm rates are paramount.

6 Conclusion and future work

This research addresses critical limitation in web attack detection through proposed BALNet (BALanced Attack Learning NETwork), a unified and reproducible pipeline that systematically incorporates ADASYN for adaptive class balancing, LDA for supervised feature extraction, and CNN based classification using the CIC-IDS 2017 [9] dataset. Extensive ablation studies systematically evaluated ADASYN against SMOTE, Random Oversampling (ROS), and Random Undersampling (RUS), feature extraction techniques and classification models including CNN, DNN, and LSTM. The results consistently demonstrate that the ADASYN-LDA-CNN configuration delivers superior robustness, scalability, and generalization.

Performance highlights encompass:

- The preprocessed pipeline of ADASYN + LDA establishes the backbone of BALNet, establishing a robust benchmark for multi-class detection.
- ADASYN emerged as superior for mitigating extreme class imbalances, by adaptively synthesizing samples in under-represented regions. In the context of web attacks, requiring approximately 0.96 s, yields a dataset characterized by a high degree of heterogeneity.

- CNN exhibited exemplary generalization capabilities with 99.87% training, 99.79% testing, 97.31% validation accuracy; losses of 0.0175/0.118), characterized by smooth convergence and negligible overfitting. DNN achieved a training accuracy of 99.88% but displayed overfitting of 90.26% validation accuracy, 2.246 loss).
- LSTM manifested instability (79.02% validation accuracy) despite a gradual learning curve.

In the context of minority web attacks:

- SQL injections (~21 samples): CNN's 99.79% testing and 97.31% validation accuracy and minimal losses indicate effective handling of sparse classes, thereby mitigating the majority-class bias commonly observed in LSTM and DNN.
- For rare attacks (XSS): Elevated testing accuracy coupled with stable validation suggests that ADASYN-LDA significantly enhanced coverage, surpassing baseline methods
- Concerning benign traffic: Near-perfect classification (>99%) was observed across models; however, CNN's lowest validation loss guarantees the absence of false positives among minority classes.
- The confusion matrix analysis further substantiates these findings. Benign traffic (Class 0 s accurately classified by all models; however, CNN and LSTM exhibit the lowest false-positive rates, making them particularly advantageous for reducing unnecessary security alerts. For moderately represented attack classes (Brute Force and XSS), all models perform competitively, with the DNN demonstrating enhanced recall for SQL Injection. Nevertheless, CNN maintains the most favorable balance between benign traffic detection and attack identification, achieving the highest overall accuracy with minimal benign-to-attack misclassifications.

Consequently, BALNet provides an optimal accuracy-stability trade-off for real-time web intrusion detection.

Future Work will aim to extend BALNet to encompass multi-stage and hybrid attack scenarios utilizing dynamic datasets, investigate CNN-LSTM hybrid architectures, implement automated tuning, and facilitate edge deployment with a focus on latency optimization.

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Author contributions
Mudita Kohli and Indu Chhabra conceived the idea for the article. Mudita Kohli performed literature search, data analysis, experimentation, and drafted the manuscript. Indu Chhabra performed the reviewed experimentation, and critically reviewed the manuscript for intellectual content. Both authors reviewed and approved the definitive version of the manuscript.

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Data availability
The datasets generated during and/or analyzed during the current study are available in the Canadian Institute for Cybersecurity (CIC) at the University of New Brunswick (UNB). [https://www.unb.ca/cic/datasets/ids-2017.html].

Human participants and/or animals
This study does not contain any studies with human or animal subjects performed by any authors.

Clinical trial number
Not applicable.

Declarations

Ethical approval and consent to participate
No primary data was collected; standard dataset is used for study, and therefore, no ethical approval was required.

Consent for publication
Not applicable.

Competing interests
The authors declare no competing interests.

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